

PHINMA UNIVERSITY OF ILOILO
College of Information Technology Education

CAPSTONE PROJECT PROPOSAL

Custom Cake Ordering Platform with Smart Scheduling and Sales Analytics

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CHAPTER 1

1.1 Background of the Study

In today's fast-paced world, businesses across all sectors are shifting to online platforms to keep up with the growing expectations of their consumers. The food service and enterprise baking industry is no exception to this trend. As tech-savvy individuals increasingly turn to digital applications for seamless, on-demand transactions, local micro-enterprises that continue to rely on manual, analog processes face significant challenges. Traditional, non-automated methods often result in critical operational inefficiencies, reduced market reach, and an inability to meet contemporary consumer demands. This shift in buyer behavior has turned digital convenience from a luxury into an absolute necessity for survival in the food retail industry. When small stores lack a modern web presence, they silently lose customers to larger competitors who offer instant online ordering.

The underlying problem stems from how consumer data is collected and managed during initial touchpoints. In a typical digital commerce environment, data inputs are structured, validated, and processed instantaneously by backend application logic.

Conversely, traditional store structures rely heavily on human

memory and fragmented paperwork to handle transactions. This lack of automated validation means that business transactions are highly susceptible to data loss, tracking delays, and operational blind spots. For community-scale enterprises, continuing to operate within a purely analog framework creates a technical barrier that cuts them off from the modern digital marketplace, making it increasingly difficult to sustain consistent growth or build long-term customer loyalty.

C&N Bakery is a local bakery operating within a highly competitive local market, offering a wide selection of baked goods categorized into standard product groups, including various daily pastries and traditional baked items. In addition to mass retail options, the establishment specializes in bespoke, high-value custom cake orders and supports a pre-ordering framework designed for clients requiring specialized baked goods for celebrations, weddings, debuts, and milestones. Despite managing a diverse product line and catering to a growing clientele, C&N Bakery handles its entire business workflow through manual mechanisms, such as physical walk-ins, phone calls, and scattered social media messages, without any centralized digital infrastructure to sustain its operations. This lack of a unified system means information is scattered across different mediums, making daily tasks stressful and

disorganized.

When a business does not use a centralized repository to store its records, its information easily gets scattered across too many different places. At C&N Bakery, a single custom transaction often leaves a fragmented data trail across multiple communication channels, where design layouts, delivery deadlines, and payment details are recorded on completely unlinked mediums. Because these data points are isolated, the staff must manually piece together the customer's requirements before coordinating production. This approach consumes massive amounts of time and makes day-to-day operations chaotic, turning simple routine tasks into exhausting manual procedures that drain the small business's daily energy.

Managing custom cake orders is inherently more complex than selling regular pastries over the counter. A custom cake order requires capturing and tracking multiple precise design dimensions, flavor variations, and stylistic preferences simultaneously. On top of that, the bakery must carefully coordinate the exact pickup or delivery fulfillment dates to guarantee product freshness and structural stability during transport. Because C&N Bakery relies on handwritten notebooks, loose paper receipts, and unorganized chat histories, it is incredibly easy for staff to forget or misinterpret these

specific design details by the time production begins. Plus, when workers try to schedule these orders by writing them down in a physical logbook, human error often causes accidental double-bookings, overloading the kitchen with more volume than the baking team can execute.

This operational vulnerability is deeply tied to the technical concept of data mutation. When a customer outlines their custom selections through casual communication channels, that information exists as an unformatted text string. As that request moves from the front counter staff to the production area, the information is manually rewritten onto paper order tags. Every time a human manually copies, interprets, or rewrites a data string, the higher the risk of transcription mistakes occurring. Essential design attributes can easily be misread or completely forgotten during the transfer. This manual workflow creates a weak system loop where the original intent of the customer rarely matches the actual physical instructions given to the baking staff.

When the bakery relies on scattered chats and unorganized paper notes, it creates a high risk of transcription mistakes during the order-taking process. These data entry errors directly lead to incorrect specifications being sent to the kitchen. Once a product is baked with the wrong design or configuration, it

triggers a chain reaction of operational failures across the business. This workflow breakdown results in wasted raw ingredients, disrupted baking queues, and unrecoverable financial losses for the bakery, while severely damaging long-term customer trust.

The financial damage caused by these data errors is made worse by the rigid nature of the baking process itself. Unlike other food service businesses that can quickly prepare a replacement dish, a custom cake requires hours of uninterrupted preparation, cooling, structural stabilization, and detailed artistic application. If a configuration error is discovered late in the production line due to a manual data entry mistake, the entire product must often be discarded. This means the bakery loses both the expensive ingredients used and the valuable time slots in the production schedule. This disruption delays every other order waiting in the queue, creating a stressful work environment and causing a backlog of delayed fulfillment timelines that ripples through the entire business day.

The operational difficulties are further worsened by the rapidly evolving commercial ecosystem in the Philippines, marked by the widespread adoption of digital transactions and electronic banking applications. Modern consumers expect seamless transaction flows that offer immediate pricing feedback, cost

transparency, and structured access to basic business policies, combined with the convenience of settling accounts remotely. For C&N Bakery, maintaining a purely manual setup limits its ability to engage this expanding, digitally native market segment that prioritizes efficiency and remote ordering. When a customer has to wait hours for a manual message reply just to get a price quote, they often abandon the purchase entirely.

This bottleneck is a direct result of operating without an automated computational engine to handle basic pricing calculations. In a manual communication model, every customer inquiry regarding custom options forces a staff member to stop their physical kitchen tasks to manually calculate costs line-by-line. This slow process creates a massive communication delay, leaving potential buyers waiting in digital queues for simple information. In the modern local economy, where consumers are accustomed to instant digital responses, these long waiting times cause high cart abandonment rates, as buyers quickly abandon the conversion funnel to buy from larger automated competitors instead.

Beyond customer-facing constraints, the absence of a unified web platform introduces severe security risks and data management obstacles. On the user side, clients lack a secure, authenticated interface to manage their records or securely

monitor the status of their orders. For administrative personnel, operating without a secure, centralized administrative console means that confidential business information—such as historical sales records, client contact logs, and sensitive operational statistics—remains vulnerable to data corruption, unbacked physical destruction, and unauthorized exposure, making data integrity a constant corporate concern. If a single paper logbook is misplaced or stained in the kitchen, the bakery completely loses its financial records and customer histories.

This total reliance on single-copy physical storage devices exposes the business to immense operational hazards. Paper logbooks kept inside an active kitchen are constantly exposed to water damage, grease stains, and accidental loss. From an information security perspective, storing a company's entire transaction history on unsecured paper slips leaves confidential client addresses and contact numbers open to unauthorized eyes. Furthermore, physical records offer zero data recovery options; if an environmental hazard occurs in the shop, years of vital business history are permanently destroyed in minutes, leaving the management with no backup logs to reconstruct their operational data.

These combined systemic vulnerabilities generate persistent

issues for both stakeholders. Customers endure a fragmented user experience due to the lack of a centralized online catalog with dynamic filters to browse products, or an intuitive mechanism to submit custom reference visual designs. Concurrently, the production and administrative staff struggle to monitor simultaneous orders across various stages of readiness, which leads to scheduling errors and fulfillment delays. Without a visual tool to track order states, progress, or daily capacities, the small kitchen team easily becomes overwhelmed.

This communication gap creates a significant operational barrier inside the production facility. Because the front counter staff and the production team do not share a synchronized information system, they easily lose track of order priorities. The production staff has no seamless way of identifying urgent queues, and the front counter staff cannot give customers accurate updates on when their orders will be ready. This forces everyone to constantly stop their physical duties to verbally verify order statuses. This constant interruption wastes valuable time, stresses out the staff, and leads to scheduling mistakes that leave customers waiting and frustrated.

To address these operational vulnerabilities and support the digital transition of the business storefront, this study proposes the development of the *Custom Cake Ordering Platform*

with Smart Scheduling and Sales Analytics for C&N Bakery. The proposed web-based application aims to establish a centralized digital infrastructure that streamlines order acquisition, balances daily kitchen capacity limits, tracks production stages, and compiles historical business records. By replacing vulnerable, manual mechanisms with a secure, database-driven workspace, the project seeks to eliminate data transcription errors, safeguard operational data, and provide an efficient, transparent web presence tailored to the bakery's operational needs.

1.2 Statement of the Problem

General Problem Statement

The primary challenge facing C&N Bakery is its heavy reliance on manual processes, physical paper logbooks, and scattered social media chat messages. Because the business does not have a single, shared online system that updates information across operations at the same time, its data is completely split up. Crucial details like order specifications, pickup deadlines, menu availability, and customer contact information are scattered across loose notes, paper kitchen calendars, and chaotic chat histories.

Relying on this unlinked, manual setup makes it incredibly easy to lose, misread, or accidentally destroy information. Without an automated web framework to instantly capture and lock in order details when a customer checks out, the bakery lacks a single, reliable source of truth for its daily transactions. On a day-to-day basis, this broken process causes constant communication mix-ups during design consultations, costly cake assembly mistakes in the kitchen, expensive ingredient waste, and accidental scheduling overbookings. Furthermore, keeping financial records and customer details in open paper notebooks leaves the bakery vulnerable to data privacy risks and permanent file loss.

Specific Statement of the Problem

1. Customer-Intake and Ordering Inefficiencies

- **Communication Backlogs and Delayed Responses:** Customers who want to browse the menu, check flavor variations, or ask about pricing are forced to call the shop or send private messages. Because communication relies entirely on manual replies from staff, the bakery experiences severe message backlogs during holidays and weekends. Customers regularly wait hours for a basic quote, leading to frustration and lost sales.
- **Pricing Guesswork and Math Errors:** Staff figure out prices right in the middle of chatting with clients by using quick mental math or looking through loose price sheets. This lack of a structured system leads to inconsistent pricing, human errors, and a total lack of transparency for the buyer, who cannot see how changing options affects their bill.
- **Misunderstood Cake Designs and Transcription Errors:** Staff must read through long chat histories and manually copy a customer's custom design requests onto physical paper kitchen slips. This messy hand-off creates a high risk of missing or miscopying critical details. When design requests are misread, it leads to incorrect cake assemblies

in the kitchen, rejected orders, and a heavy waste of expensive baking ingredients.

- **Information Gaps and Customer Hesitation:** C&N Bakery's manual setup offers no public, instant access to storefront policies, allergen warnings, or payment guides. This lack of clear, immediate information forces customers to ask repetitive questions and causes transaction hesitation, especially when buyers are reluctant to send down-payments without seeing clear shop guidelines upfront.

2. Operational, Scheduling, and Production Disruptions

- **Kitchen Overbooking and Overlapping Deadlines:** Fulfillment deadlines are tracked using a single physical calendar in the kitchen. Because different employees take orders simultaneously over the phone and social media without a shared view of the schedule, the bakery regularly overbooks specific dates. This overloads the kitchen team, leading to stressful rushes, dropped decorating quality, and late customer pickups.
- **Disorganized Production Tracking and Status Blindness:** The kitchen staff does not have a centralized, real-time method to see where open orders stand. Instead, employees have to yell across the room to ask about progress or constantly flip through loose paper notes to find an order. This

forces the bakers to stop their detailed cake decorating tasks just to answer status questions from the front counter.

- **Logistical Misalignments and Delivery Mix-ups:** Because pickup and delivery criteria are written down on loose paper documentation, it is incredibly easy for staff to misread a date or mix up order notes. A worker might accidentally prepare a delicate custom cake days before the customer actually wants it, or miss a deadline entirely, creating chaotic delivery mistakes.
- **Slow Hand-offs and Manual Record Keeping:** Confirming that a sale was successfully finished requires manual follow-ups and physical file logging. If a tired worker forgets to file a completed order slip into the physical ledger book at the end of an exhausting shift, that entire sales record effectively vanishes from the day's bookkeeping, creating financial blind spots.

3. Data privacy and bookkeeping risks

- **Unsecured Business Records and Lack of User Privacy:**
Operating without secure, role-based database access leaves C&N Bakery open to data privacy risks. Customers lack a private interface where they can safely access their accounts, while confidential business financial logs and

customer databases are kept on unbacked paper notebooks or personal mobile phones, leaving sensitive business data exposed to physical theft, damage, or unauthorized exposure.

- **Slow Payment Tracking and Accounting Errors:** Managing down-payments and cash balances using manual notebooks is highly labor-intensive and prone to math mistakes. Because proof-of-payment screenshots from digital wallets are kept loose in mobile phone photo galleries, matching a payment slip to a specific paper order requires tedious manual cross-checking, causing frequent money discrepancies.
- **Absence of Business Insights and Data Blindness:** The bakery owner cannot easily check if the business is growing or calculate profit margins without spending hours doing manual math at the end of every month. To see how the bakery is doing, the owner must manually sort through and add up hundreds of individual physical order slips line-by-line, leaving management unable to track changing customer preferences or predict busy seasonal rushes.

1.3 Objectives of the Study

General Objective

The primary objective of this study is to design, develop, and implement a secure, web-based ordering, smart scheduling, and sales analytics platform for C&N Bakery to transition its manual transactional workflows into a shared online platform. This study aims to replace paper-dependent habits—such as manual logbooks, physical kitchen calendars, and fragmented communication threads—with an integrated platform that shares data access across customers, kitchen personnel, and management. Ultimately, the system seeks to eliminate pricing ambiguities through automated cost computation, protect the kitchen from operational overbooking via automated scheduling guards, and provide business analytics to support data-driven decision-making.

Specific Objectives

To successfully achieve the general objective of the study, the research will fulfill the following structured engineering and research milestones:

1. To analyze and document the existing manual workflows of C&N Bakery regarding product ordering, custom cake configuration, kitchen scheduling, and sales record-keeping to establish the foundational technical requirements of the platform.
2. To plan out how the website will work, map out the database

where information is stored, and design clean, easy-to-use screens. This ensures that the shop's data stays accurate, sensitive business details remain password-protected, and the website is simple for everyone to navigate.

3. To build the public website that customers see, combining an online menu that lets users filter pastries by category with a custom cake designer page. This designer setup will automatically calculate and update the estimated cake price on the screen the moment a customer changes their design choices.
4. To build a secure checkout step on the website that lets buyers send their custom text messages and upload design reference photos along with their order. This guarantees the customer's design requests travel safely straight to the shop's database without getting lost or altered.
5. To build an automated scheduling tool that keeps track of how many custom cake orders are placed for a specific date. Once the kitchen hits its daily limit, the system will automatically lock that calendar date on the website so no more customer bookings can be made for that day.
6. To build an order tracking layout for the kitchen staff that displays all of a customer's cake choices, text notes, and uploaded photos together on a single view. This allows the baking team to see all specifications at once and

systematically update an order's progress as it moves from baking to final decorating.

7. To build a reporting section for the owner that automatically gathers past sales records and displays them in clear visual charts. This makes it easy for management to monitor monthly revenue trends and instantly see which menu items are selling the best.
8. To set up secure user login rules and data validation steps across the website. This ensures that customers and staff only see the specific information they are allowed to access, while protecting private business financial logs and preventing typing errors during checkout.
9. To test and evaluate the completed website through structured trial runs with developers and actual users. This process measures how fast the system runs, verifies that the security rules work correctly, and uses standard evaluation feedback to ensure the website is easy and practical for everyone to use.

1.4 Scope and Delimitation

This study focuses on designing, building, and testing a secure website for ordering, smart scheduling, and sales analytics tailored specifically for C&N Bakery. The goal of the system is

to move the bakery's manual, paper-dependent habits—like handwritten notebooks and scattered social media messages—into a single, organized online platform. The website is split into a public customer portal and a secure management dashboard to streamline operations.

1. System Environment and Accessibility

The platform is built exclusively as a responsive web application, so users do not need to install a mobile app. The system uses a flexible grid layout that automatically adjusts its screens to look and work great on desktop computers, laptops, tablets, and smartphones. The website works across modern web browsers—such as Google Chrome, Mozilla Firefox, Apple Safari, and Microsoft Edge—as long as the user has an active internet connection.

2. User Roles and Access Levels

To keep data accurate and secure, the system splits user access into three distinct roles:

- **Bakery Customers:** Public visitors can browse the digital product menu and view store policies. Registered and logged-in buyers get a private account space to set up custom cake requests, enter order details, upload photos of their digital payment receipts, and track the progress of

their active orders.

- **Production Staff:** Logged-in kitchen staff and decorators are sent straight to the production dashboard. They can view complete design instructions and upload reference photos to advance orders through different baking stages. Production staff are blocked from seeing financial data or changing product prices.
- **Bakery Owner (System Administrator):** The owner has full access to the entire backend system. They have exclusive controls to manage user profiles, change calendar availability, update menu inventories, adjust base pricing, control which items are visible on the shop, and view automated business analytics reports.

3. Scope of the Customer Portal

The customer website includes all the features needed to guide a buyer from browsing the menu to submitting an order:

- **Categorized product catalog:** The shop organizes items into clear product sections with filtering options, helping users find pastries and local treats quickly.
- **Live price calculator:** This page guides customers step-by-step through custom requests. As users pick their options—like tiers, sizes, flavors, and frosting—a

background script automatically calculates and updates the estimated baseline price on the screen so pricing is completely transparent.

- **Design reference uploader:** The order form lets customers attach inspiration photos and type out cake dedication text directly into the system, preventing design and spelling mistakes.
- **Order limit checker:** During checkout, the customer chooses a fulfillment date on an interactive calendar. Dates that have hit the bakery's maximum daily limit are automatically greyed out by a background scheduling rule. The form then collects necessary pickup or delivery details.
- **Self-Service Order Progress Tracker:** Logged-in users can check a status tracker in their profile to see exactly what stage their cake is in as the kitchen updates it, eliminating the need to call or text the shop for updates.

4. Scope of the Administrative and Kitchen Dashboard

The secure backend dashboard gives management and the kitchen team full control over daily workflows, schedules, and records:

- **Bakery performance summary:** Upon logging in, the owner sees an overview of key performance indicators, like total revenue for the month, pending tasks, and active kitchen

workloads.

- **Main order directory:** This section groups all incoming orders into a clean, searchable directory. Staff can filter records by customer details, product types, deadlines, and payment status without touching paper.
- **All in one design summary:** To avoid mistakes in the kitchen, clicking an active order opens a single layout displaying the customer's uploaded reference image alongside their exact flavor choices and text notes.
- **Step-by-Step Production Status Controller:** The kitchen view lets bakers smoothly update an order's status as it moves through different preparation stages. Staff can also check and approve payment status after reviewing customer receipts.
- **Instant menu toggles:** Staff can turn off the visibility of specific menu items on the backend, which removes out-of-stock pastries from the public storefront instantly during ingredient shortages.
- **Auto booking limiter:** This tool runs continuously in the background to count incoming orders for each date. The moment a daily limit is met, the system automatically blocks that date from accepting more online reservations.
- **Sales Analytics Visualizer:** This administrative tool

gathers historical transaction records from the database and turns raw numbers into clear charts, highlighting monthly revenue trends and the bakery's most popular products.

5. Delimitation of the Study

To keep the project realistic and ensure it is finished on time, the following features are explicitly excluded from this study:

- **No Native Mobile Applications:** The project is strictly confined to a responsive web browser design; it does not include developing or publishing standalone mobile apps for Google Play or the Apple App Store.
- **No Direct Financial API Integrations:** The website does not connect directly to banking systems or automated payment gateways. Payment verification is handled manually by the staff, who review uploaded receipt screenshots and manually mark the order as paid.
- **No Automated Third-Party Logistics Routing:** The application works completely independently of external courier networks. It does not link with delivery services (like Grab or Lalamove); setting up drivers and managing delivery dispatch must be done by the staff outside the system.
- **No Raw Material Inventory Tracking:** The system does not

track raw baking ingredients (like the weight of remaining flour, sugar, or eggs). Inventory management relies entirely on staff observations and using the manual storefront visibility toggles.

- **No Human Resource or Payroll Systems:** The platform is built solely for orders and scheduling. It does not include employee timecards, shift scheduling, or payroll management features.
- **No Supply Chain or Vendor Logs:** The software lacks tools for managing raw ingredient suppliers, creating wholesale purchase orders, or tracking fluctuating ingredient costs from vendors.
- **No Offline Operational Capability:** The application depends completely on a live database server. It does not support offline mode or offline data saving; users must have a working internet connection to use the website.

1.5 Significance of the Study

Building the Custom Cake Ordering Platform with Smart Scheduling and Sales Analytics brings great practical value to everyone involved in the daily operations and growth of C&N Bakery. As

the local food and bakery industry shifts toward digital spaces, traditional neighborhood bakeshops must update their tools to stay relevant and keep up with modern competitors. This study serves as a real-world demonstration of how a local bakery can successfully transition away from the stress and physical disorganization of manual paperwork. By replacing easily lost handwritten logbooks, fading sticky notes, and scattered social media message threads with a single, all-in-one web application, this project creates a smooth, dependable ordering process. This new system ensures absolute pricing honesty for everyday customers, protects the kitchen team from accidental booking overlaps, and safeguards historical business records from getting physically damaged or lost. By modernizing these daily tasks, this study provides clear, long-term benefits to a wide group of stakeholders:

1. Bakery Customers and Guests (The Buyers)

Traditionally, everyday buyers trying to order custom cakes or local delicacies from neighborhood shops face frustrating customer service ordering roadblocks, including scattered chat histories, delayed responses, and unclear pricing structures. The launch of this online platform directly assists buyers by providing:

- **Honest and Instant Pricing:** Customers benefit from complete price transparency, receiving real-time cost feedback as they select different cake configurations and design complexities, which removes pricing guesswork and transactional anxiety.
- **Accurate Design Submission:** The platform allows buyers to seamlessly submit their design reference photos alongside their flavor preferences and structural specifications, minimizing design misunderstandings and ensuring their expectations match the final product.
- **Independent Order Tracking:** Users gain access to real-time status updates regarding their active orders directly from the bakery, eliminating the need to wait for manual staff replies to confirm the current production phase.
- **Convenient Menu Browsing:** The centralized digital catalog allows customers to comfortably browse offerings and plan their celebrations at any time from any internet-connected device.

2. Administrative Control Personnel (Bakery Staff and Decorators)

Kitchen staff and cake decorators working within a manual setup operate under immense daily pressure to track upcoming deadlines and unique design themes using fragile paper records. This

project introduces organization to the kitchen workspace by providing:

- **Workload Protection:** The platform prevents accidental overbooking by regulating the daily volume of custom orders, protecting the kitchen team from exhausting workloads and operational burnout.
- **Production Accuracy:** Cake decorators can review complete design instructions and reference images within a single, unified view, eliminating errors caused by searching through loose papers or disjointed chat threads.
- **Step-by-step production stages:** The kitchen interface allows baking personnel to smoothly advance orders through verified preparation milestones, moving a cake from mixing to baking and final decorating.
- **Efficient Menu Management:** Staff can update product availability instantly, preventing customers from ordering and paying for treats that are currently out of stock due to raw ingredient shortages.

3. Executive Management (The Bakery Owner)

For the business owner, managing a specialized shop without a centralized digital tool complicates financial auditing and long-term business planning. This platform streamlines daily

oversight by providing:

- **Automated Financial Records:** The system organizes transaction histories, down-payments, and payment verifications into a reliable digital ledger, reducing the risk of human calculation errors.
- **Visual Sales Insights:** The owner gains access to automated performance tools that track long-term business growth and highlight top-selling products, facilitating data-driven inventory and financial planning.
- **Data Protection and Privacy:** Secure role-based access rules ensure that sensitive financial metrics and customer details are restricted solely to the owner, keeping business records safe from unauthorized eyes.
- **Strategic Business Decisions:** By providing unalterable digital summaries of customer buying trends, the platform helps the owner make smarter choices regarding ingredient budgeting and seasonal menu updates.

4. The Organization (C&N Bakery as a Business)

Transitioning to an automated platform professionalizes the C&N Bakery brand and protects its commercial reputation from the fallout of missed deadlines or forgotten details. The business benefits from:

- **Prevention of Revenue and Resource Waste:** Enforcing systematic confirmation and tracking parameters ensures the kitchen never wastes costly ingredients on unverified custom orders.
- **Stronger Local Competitiveness:** The platform grants a small, family-owned shop a massive competitive edge by offering the modern digital conveniences that tech-savvy buyers expect today.
- **Scalable Business Growth:** By streamlining internal workflows and automating communications, the bakery can safely handle higher order volumes during peak holiday seasons without overextending its administrative capabilities.

5. Future Systems Developers

For software engineers building e-commerce applications for specialized food markets, standard online stores lack the strict booking limits and custom configuration tools required for custom baking. This research assists developers by offering:

- **Custom order reference:** This study serves as a practical guide for building custom product forms that calculate prices in real time as users click different options.
- **Booking Rules Framework:** This study provides a practical

template for setting up automated calendar boundaries that prevent system overbooking based on fixed daily shop capacities.

6. Future Academic Researchers

This study provides valuable documentation for researchers analyzing how small businesses transition from manual operations to localized web platforms. It provides documented local business workflows that can be directly compared to other retail models and serves as a foundational reference for designing systems for made-to-order businesses. Furthermore, it outlines structured paths for future student projects, such as integrating automated SMS notifications or third-party logistics applications.

7. The Local Food Service Industry

The success of this platform demonstrates that practical, high-quality digital solutions are accessible to neighborhood bakeshops and small restaurants without requiring massive corporate budgets. Raising local service standards through transparent pricing and digital reliability encourages nearby food vendors to modernize their operations, elevating the industry as a whole.

8. BSIT Students

For Information Technology students, this documentation serves as a practical blueprint for breaking down manual business problems into functional technical features. It demonstrates how to translate real-world storefront operations into clean database states and systematic workflows, illustrating the core principles of effective user interface design and role-based data security.

1.6 Definition of Terms

To ensure a clear and accurate understanding of the technical concepts, system operations, and structural components discussed throughout this study, the following key terms are defined both conceptually (how they operate in general) and operationally (how they are applied inside the C&N Bakery platform):

1. Web-Based System

- **Conceptual Definition:** An application software architecture that runs on a remote server and is accessed exclusively via a standard web browser over an internet connection, requiring no local installation or hard drive storage on the user's local machine.
- **Operational Definition:** This refers to the online software platform engineered specifically for C&N Bakery. It allows customers, cake decorators, and the business owner to access the retail menu, place cake pre-orders, monitor tracking statuses, and evaluate monthly financial metrics using any available internet-connected device—such as a smartphone, tablet, laptop, or desktop computer—without having to download an application file from an app store.

2. User Authentication and Account

Access Control

- **Conceptual Definition:** A digital security gatekeeping protocol that verifies a user's identity through login credentials and checks their pre-assigned database permissions before granting access to protected sections of a network.

- **Operational Definition:** This functions as the system's security checkpoint during the login sequence. By reading the user's account type, the system automatically routes the user to their appropriate environment. This ensures that everyday customers are restricted to the public ordering pages, kitchen staff are directed only to production workflows, and the bakery owner maintains exclusive, password-locked access to sensitive financial sales records.

3. Interactive Custom Cake Builder

- **Conceptual Definition:** A dynamic front-end web portal interface that allows customers to visually select, configure, and modify specific design elements of a product prior to finalizing a transaction.
- **Operational Definition:** This is the dedicated configuration page on the website where customers design a custom cake from scratch. Users can select a guided sequence of design variables—including the total number of tiers, layer sizes, sponge flavors, and frosting variations—while simultaneously typing written text inscriptions and uploading visual design reference photos.

4. Automated Pricing Calculator

- **Conceptual Definition:** A client-side computing script that runs in the background of an interface to instantaneously compute arithmetic totals the exact millisecond a user alters an input variable on the screen.
- **Operational Definition:** This is the mathematical background script operating inside the Interactive Custom Cake Builder page. The instant a customer changes a design option, such as upgrading a cake tier size or picking a premium frosting flavor, the script recalculates the baseline cost metrics and updates the final price quote on the user's screen in real time, removing the need for manual price quotes via phone messages.

5. Online Menu Catalog

- **Conceptual Definition:** A structured digital storefront layout that organizes a commercial business inventory into searchable, filterable categories to ease online navigation.
- **Operational Definition:** This serves as the public digital browsing showcase for C&N Bakery's full baked product line. It automatically separates items into distinct product category views—such as customized cakes, cupcakes, kakanin,

and sapin-sapin—and provides a text-based search input alongside quick budget filters so users can find specific treats that fit their spending limits.

6. Private Customer Dashboard and Progress Tracker

- **Conceptual Definition:** A restricted user profile workspace that displays personalized account data, transaction history lists, and real-time operational status updates regarding active requests.
- **Operational Definition:** This is the private account hub accessible only to registered and logged-in C&N Bakery customers. Within this workspace, users can safely manage their account details, view past transaction records, and monitor a live visual timeline that reads data directly from the kitchen to show if their active order is currently marked as Preparing, Baking, Packing, or out for Delivery.

7. Sales Analytics Charts

- **Conceptual Definition:** A data visualization module that parses archived transaction logs from a relational database and charts them into graphical formats to highlight operational trends.
- **Operational Definition:** This reporting system automatically

gathers completed order histories and transforms the raw numbers into visual data tools: a Monthly Revenue Graph that tracks long-term business growth, and a Top-Selling Products Graph that visually highlights which items, such as traditional kakanin or custom birthday cakes, are generating the most revenue.

8. Menu Availability Toggle Control

- **Conceptual Definition:** A backend binary administrative switch used by system operators to instantly modify the visibility status of a product row within a public-facing database catalog.
- **Operational Definition:** This refers to the digital availability controls positioned next to every individual pastry listing in the admin management panel. If the kitchen runs out of the raw ingredients required to produce a specific item like sapin-sapin, a staff member can adjust the setting, which instantly hides that item from the public storefront to prevent accidental orders for sold-out treats.

9. Auto lock booking calendar

- **Conceptual Definition:** A data-driven scheduling guard that tracks daily reservation frequencies against a

predetermined threshold and automatically locks calendar dates from receiving further entries once the limit is reached.

- **Operational Definition:** This is the system's background workload gatekeeper. It continuously counts incoming cake reservations assigned to a specific date and automatically locks that date on the customer's booking calendar the exact moment the day hits its maximum management-defined threshold of active custom cake orders, protecting the kitchen team from accidental overbooking.

CHAPTER 2

2.1 Foreign Studies

Shafiei, Khonsari, and Mousavi (2022) confirm that transitioning from fragmented, traditional hardware-bound server architectures to a dedicated serverless computing framework significantly improves system management, resource efficiency, and software deployment speed. Their research emphasizes that moving away

from monolithic, legacy setups to a centralized cloud architecture allows for the automated handling of computational infrastructure, which prevents the heavy operational maintenance, constant framework upgrades, and manual operating system patching often associated with running physical, on-site hardware. The system paradigm was examined across multi-dimensional modules, exploring how serverless spaces utilize Function-as-a-Service (FaaS) and Backend-as-a-Service (BaaS) ecosystems to cleanly isolate application business logic inside cloud-level micro-containers, allowing development teams to focus entirely on running code smoothly without the low-level administrative burden of configuration tuning or security sandboxing.

The study's technical evaluation phase involved an in-depth analysis of dynamic scalability modeling across parallel and distributed workloads, focusing heavily on execution latency bottlenecks like "cold starts"—which occur when an idle function is triggered after a long resting period—and evaluating modern optimization techniques, such as keeping container images warm or running lightweight pre-warmed background micro-runtimes to reduce initial response delays.

Furthermore, the researchers dive deep into the specific operational realities of containerization, documenting how

traditional server environments inherently trap application code inside fixed operating systems that require ongoing human oversight, while serverless frameworks shift machine management, security isolate sandboxing, physical hardware upkeep, and underlying database maintenance entirely to the service provider. This structural decoupling removes the low-level administrative burden from developers, reducing architectural friction and minimizing software vulnerabilities caused by mismatched system dependencies or delayed server patches. The authors detail the dynamic scalability mechanisms of cloud environments, explaining that serverless platforms provide high speed and inherent dynamic scalability to efficiently handle highly dynamic, parallel, and distributed workloads. When an application experiences a massive wave of incoming traffic, the cloud framework executes parallel computing tasks by instantly spinning up isolated micro-container runtime instances, which successfully avoids the CPU exhaustion, memory leakage, and request queue congestion that typically freeze traditional servers during traffic surges.

Later, when traffic volumes drop and the system returns to a resting state, the serverless framework optimizes financial resource utilization by automatically spinning down the idle micro-containers, preventing unnecessary server runtimes.

Additionally, the researchers provide a detailed evaluation of the pricing issue and financial structures found in serverless deployment spaces, highlighting the value of fine-grained cost models compared to traditional hosting paradigms. Traditional hosting web backends lock small operations into fixed, recurring leasing fees based on projected peak usage limits, meaning organizations pay full price for computational infrastructure even during slow hours when their applications are idle.

Serverless monetization models, by contrast, operate on a highly granular pay-as-you-go scale that measures performance by calculating execution time down to the exact millisecond alongside the specific memory size used per query. This ensures that when an application is sitting idle, operational expenses drop to zero.

The study concludes that the serverless platform provides clear architectural advantages for application developers looking to eliminate local infrastructure management, making it an incredibly robust, secure, and cost-efficient environment for deploying modern, highly interactive web systems.

This study is highly relevant to the development of the C&N Bakery platform because it provides the technical reasons for using a serverless system. By using Google Firebase as our Backend-as-a-Service (BaaS) system, the web application does not

need a local physical server setup. This completely removes the hard work of building and fixing server hardware, which perfectly aligns with the authors' points about reducing operational complexity. This setup is highly practical for a small, local business like C&N Bakery because they do not have an on-site IT department or a dedicated database team to look after heavy hardware. By running everything through Firebase's cloud services, the system stays online 24/7 without the physical risks and security problems that come with old-fashioned server setups.

The authors' findings on how serverless systems handle big spikes in web traffic directly solve a major problem for C&N Bakery. The bakery experiences massive changes in order volumes throughout the year. They have normal, quiet traffic on regular weekdays, but they get hit with an overwhelming number of custom cake orders and image uploads during major holidays like Christmas, New Year, Valentine's Day, and Mother's Day. A normal web host might crash if too many customers try to customize cakes and check out at the exact same time. By using the serverless auto-scaling feature proven in this study, our platform can handle these busy holiday rushes smoothly by instantly pulling in more cloud power to keep the pages loading fast and ensure no customer's order or payment gets lost.

Lastly, the pay-as-you-go cost model discussed in the study makes the C&N Bakery platform highly affordable and sustainable. Traditional web hosting forces a business to pay an expensive, flat monthly fee even if nobody is using the website. Firebase, on the other hand, only charges the bakery for the exact amount of database reads, writes, and file transfers that happen when people actually use the application. This means during quiet business weeks, the system costs almost nothing to maintain.

Furthermore, the study's section on event-driven functions explains how our automatic emails work: the moment a baker updates an order's progress on the dashboard, a background cloud function instantly triggers and sends an email receipt to the customer's inbox without slowing down the rest of the website.

Sulova (2023) confirms that transitioning from fragmented, legacy digital storefronts to a comprehensively integrated and technologically advanced e-commerce framework significantly improves operational performance, data consistency, and organizational scalability for small and medium-sized business networks. The research emphasizes that a centralized, multi-layered system integration methodology prevents the

information loss, manual administrative bottlenecks, and data disjointedness often associated with running disconnected operational tools and loose transactional systems. The conceptual framework was examined across precise core developmental directions, exploring the application of modern software tools, the enhancement of the foundational database base, and the dynamic possibilities of total architectural connectivity. The study's technical evaluation phase utilized a highly rigorous research methodology, leveraging advanced computer natural language processing and empirical text-mining analysis via platforms like RapidMiner Studio to systematically process a vast repository of scientific literature and isolate the core performance drivers of electronic retail performance.

The findings showed that e-commerce optimization must be executed across three explicit levels of connectivity: application integration, data-level integration, and external partner system integration. At the application integration level, the researcher documents how modern web stores must directly bundle primary user screens with auxiliary automated components, such as automated messaging paths, custom recommendation loops, and customer communication workflows. At the data-level integration stage, the study argues extensively for the deployment of a unified database infrastructure that

actively merges unstructured transactional data streams, successfully eliminating isolated informational silos and ensuring that any data-driven event occurring on the frontend user portal maps directly to backend tracking tools in real time. To achieve true systemic unity, the paper highlights that database entries should not exist as single, unlinked rows but should actively communicate across every administrative view. This structural integration allows operational fields—such as order state transitions, real-time balance metrics, and customer contact parameters—to be updated instantly across the entire application ecosystem when a target field is toggled. By binding frontend inputs to structured backend cloud parameters, the architecture ensures absolute data alignment and complete software resilience against system failures.

Furthermore, the author deeply investigates how system integration can completely change how businesses handle their records. The study breaks down how manually moving data from one software to another introduces heavy human error and tires out business staff. Sulova outlines a system layout where transaction histories are parsed instantly by automated cloud codes. This background process works day and night, evaluating item categories and checking totals without requiring a person to type anything into an Excel sheet. By designing automated

pathways that constantly clean up and calculate incoming order metrics, the application can host active visual dashboards that show business charts instantly. This gives managers immediate access to live updates, allowing them to spot changing consumer habits and tweak their selling plans right away. Ultimately, the researcher proved that pairing client-facing shopping sites with unified backend management workspaces drastically cuts down the time spent on administrative tasks, removes clerical mistakes, increases tracking clarity, and hands small businesses the professional tools they need to grow and succeed in highly competitive digital markets.

This study is highly relevant to our project because it provides the practical blueprint we need to build the system design for the C&N Bakery platform. Our application directly solves the problem of scattered data by merging the customer menu catalog, the 4-choice cake customizer, the smart scheduling calendar, and the manager's dashboard into a single, unified website. This matches Sulova's core concept of removing disconnected software silos. By putting all these features into one codebase, we can eliminate the bakery's old reliance on physical paper books and messy social media chat logs. This clean digital environment makes managing orders simple and effectively cuts the bakery's historical manual data entry mistakes down to zero.

The study's focus on connecting frontend user screens directly to live backend databases supports our technical programming choices. When a customer uses the C&N Bakery storefront, the system uses JavaScript to instantly package the custom design parameters from the cake order form into a clean JSON data structure. This data is pushed immediately into a serverless NoSQL database. Because the database is fully integrated with the manager's screen, the order details pop up on the kitchen dashboard right away without requiring the page to be manually reloaded. This real-time synchronization prevents lost orders and ensures that the kitchen staff can see exactly what options the customer chose when the second checkout is completed.

Additionally, the study's section on automated data integration supports the development of our Revenue Management Canvas.

Instead of forcing the bakery owner to spend hours reading through manual pen-and-paper journals at the end of the month, our platform uses background calculation functions to automatically read the transaction history. The system changes raw database entries into easy-to-read charts, showing vertical monthly sales bars and horizontal graphs of top-selling products. This gives the owner clear visual tools to track daily growth and see which cake flavors or sizes generate the highest profit margins, completely removing the risk of manual math

errors. Furthermore, this structural dashboard update helps the owner stay blind-spot free, allowing management to track busy customer trends, prepare the right amount of ingredients, and plan for long-term expansion using real business facts instead of pure guesswork.

Pereira and do Prado (2011) confirm that transitioning from fragmented, rigid software engineering methods to a dedicated agile development framework significantly improves requirements engineering, user workflow alignment, and design flexibility for web applications. Their research emphasizes that an iterative development process prevents project delays, communication gaps, and software failures often associated with large upfront

manuals and static specification sheets. The agile process was examined across three distinct disciplines, exploring how development teams can systematically refine user stories by pairing them with a structured Navigation Model and physical Story Cards to guide application coding. The study's technical evaluation phase utilized a highly rigorous methodology, tracking requirement lifecycles across collaborative web engineering paradigms, Behavior Driven Development, Test Driven Development, Scrum, and eXtreme Programming to achieve high real-world system adoption rates. The findings showed that breaking software requirements down using the 3 C's principle (Card, Conversation, and Confirmation) and the INVEST criteria accelerates construction by focusing strictly on features that bring direct value to the end-user.

At the planning level, the researchers document how a spatial Navigation Model allows developers to map out user paths and track exactly how information flows between separate user interface panels before a single line of backend logic is written, uncovering hidden interaction gaps or broken pathways early in the development lifecycle. At the technical validation stage, the study argues extensively that using Story Cards enforces clear software boundaries by establishing concrete data fields and validation rules directly based on how workers

perform their daily tasks.

Furthermore, the authors deeply investigate how running continuous feedback loops and user conversations improves the design of interactive user interfaces, allowing developers to optimize data fields and functional modules to match changing client workflows. By implementing this highly participative approach, the researchers concluded that the system not only dropped administrative friction and lowered software development costs but also increased overall system trust and customer satisfaction by delivering a digital product that smoothly handles real-world business routines.

This study is highly relevant to our project because it provides a strong justification for using an Agile development lifecycle to build the C&N Bakery platform. Because a small bakery's operational setups and menu options can easily change, using an iterative sprint model allowed us to adjust the website's features as the bakery owner and staff tested early versions. This collaborative approach prevented project delays and kept us from building features the business did not actually need. To keep development organized, we divided the project into three distinct sprint windows: Sprint 1 delivered the public storefront with its price-bracket sorting canvas (P0-P199, P200-P500, and P501+); Sprint 2 focused on the core manager

workspace, the 4-stage kitchen order tracker, and the split-screen image layout; and Sprint 3 tied everything together by integrating the automated analytics and email systems.

The study's focus on gathering requirements based on real-world business constraints supports our field data collection strategy. Instead of making generic assumptions about how an e-commerce store should work, our team spent time inside C&N Bakery's kitchen to map out their exact manual routines and physical capacity limits. This hands-on analysis gave us the precise rules needed to program smart software constraints. For example, we used these findings to build the Smart Scheduling Matrix Calendar, which automatically blocks out specific pickup dates on the customer checkout screen the moment a day hits its maximum capacity limit of three (3) custom cake orders.

Finally, the validation rules discussed in the paper guided our User Acceptance Testing (UAT) phase. By testing the live website alongside the bakery owner, bakers, counter clerks, and ten regular retail customers, we gathered direct, practical feedback on how the system performs under real working conditions. This step helped us fine-tune the Split-Screen Layout Blueprint inside the order details window. By listening to the staff's input, we ensured that the cake decorators can clearly view custom design notes and customer reference photos side-by-side

on a single screen, preventing eye strain and eliminating any confusion during busy kitchen production shifts.

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2.2 Local Studies

Abellana and Alonzo (2025) confirm that transitioning from fragmented, manual transaction methods to a dedicated, automated digital framework significantly improves operational efficiency, revenue protection, and business growth rates for micro, small, and medium-sized bakery enterprises in the Philippines. Their research emphasizes that relying on traditional, open-counter selling and inconsistent pen-and-paper record-keeping routines

leads to data loss, chaotic ingredient tracking, and a total inability to monitor real business growth patterns. The system boundaries of local baking operations were examined using a highly rigorous descriptive-quantitative research methodology that systematically analyzed the operational realities and distribution pathways of 113 small independent bakeries across regional economic environments. The findings showed that entrepreneurial human capital—specifically the traits of business agility, operational adaptability, and technological complementarity—exerts a highly statistically significant and positive impact on an enterprise's actual survival rate, while the modernization of distribution channels serves as an essential structural driver for scaling business operations.

At the administrative level, the researchers document how external disruptions and physical mobility limitations severely damage traditional, face-to-face baking models, forcing business owners to actively throw away manual logbooks that frequently get lost, broken, or misread. The study notes that under traditional methods, bakery managers remain completely unable to plan daily baking batches or calculate order balances effectively, especially when dealing with fluctuating market prices for raw ingredients like flour, sugar, and yeast. To resolve these structural operational roadblocks, the research

places special focus on a complete transition toward digital transformation readiness, explicitly proving that combining automated management tools with digital storefront channels drops administrative friction and completely eliminates human tracking errors.

Furthermore, the authors deeply explore the broader socioeconomic difficulties that local bakeries face when managing day-to-day work tasks during unexpected economic shifts. The study expands on the fact that small bread and pastry shops operate on thin profit margins, meaning that even minor mistakes—like overbaking slow-selling goods or forgetting a customer's advanced deposit—can quickly ruin a shop's weekly cash flow. Abellana and Alonzo detail how introducing localized technology gives managers a direct look into their actual performance trends, showing them exactly which products move fast and which ones sit on shelves too long. By removing human guesswork and setting up a secure, unfragmented digital pipeline, local food producers can smoothly adjust their daily baking schedules to match real market demands. Ultimately, the researchers concluded that moving away from old-fashioned open-counter setups not only protects small businesses from losing track of their sales but also builds long-term operational resilience, allowing local bakeries to

professionalize their operations and safeguard their earnings in a highly competitive local market ecosystem.

This local study is highly relevant to our project because it provides direct, real-world proof of the exact business struggles currently facing C&N Bakery. Just like the 113 bakeries surveyed in Mindanao, C&N Bakery operates as a small-scale pastry enterprise that is held back by old-fashioned paper logs and face-to-face ordering habits. The authors' proof that inconsistent paper record-keeping stops a bakery owner from tracking real business growth perfectly justifies why our project is throwing away C&N Bakery's manual logbooks and replacing them with a centralized, automated web dashboard.

The study's findings on the need for "modernizing distribution channels" directly support our design choices for the public storefront. C&N Bakery cannot grow if it only relies on walk-in customers and physical menus. Our web platform addresses this limitation by giving the bakery a complete online storefront. This allows customers to browse products, check pricing tiers via our sorting canvas (~~P0-P199~~, ~~P200-P500~~, and ~~P501+~~), and send order details from their mobile devices or computers. This system creates a clean, digital distribution loop that removes the physical barriers highlighted in the study.

Finally, our choice to link this storefront to a serverless

Google Firebase backend answers the authors' call for "automated systems and digital transformation." Instead of the bakery staff spending hours listing sales numbers by hand, our database system works quietly in the background. It records transactions automatically and instantly feeds them into our visual data tools, such as the vertical monthly sales charts. This integration gives C&N Bakery an efficient way to manage order pipelines, calculate its true daily revenues, and avoid financial tracking errors without needing complex technical training.

Ballena, Caday, Iman, and Rolle (2022) confirm that transitioning from fragmented, unstructured social media messaging setups to a dedicated, structured web-based ordering framework significantly improves operational consistency, minimizes financial losses from communication errors, and reduces administrative burnout for micro-scale baking entrepreneurs in the Philippines. Their research emphasizes that relying on informal, loose chat logs for custom pastry

transactions leads to chaotic information-gathering loops, missed structural details, and expensive production mistakes that ruin small business cash flows. The operational boundaries and lifecycles of local baking shops were examined using a basic interpretive qualitative research design paired with purposeful sampling, allowing the investigators to run systematic thematic extraction via open, axial, and selective coding to map the actual working realities of small-scale cake and bread sellers. The findings showed that while commercial social media platforms provide a free marketing channel to reach local consumer groups, they introduce severe data processing bottlenecks over time that actively stall long-term financial growth and strain small business systems.

At the production level, the researchers document how independent cake makers often work as one-person operations where a single individual must handle raw ingredient sanitation, dough mixing, oven runtime monitoring, intricate icing decoration, customer messaging, and courier rider coordination simultaneously. The study notes that under traditional, chat-based methods, bakers experience extreme physical and mental stress during holiday peaks and weekend celebration rushes because they are forced to spend hours answering disjointed chat notifications instead of focusing on maintaining

pastry quality and food safety tracking. To resolve these structural operational roadblocks, the research places special focus on moving away from unstructured messaging environments, explicitly proving that standardizing consumer transaction paths through a centralized, data-locked system dramatically lowers administrative friction and prevents the human clerical errors that frequently lead to costly product waste. By implementing this fixed digital pipeline, the researchers concluded that the system not only protected customer brand loyalty but also increased overall kitchen safety and business sustainability, delivering an efficient digital environment where incoming transactions instantly convert into organized, error-free kitchen instructions the moment an online order is cleared.

Furthermore, the authors deeply evaluate how these manual communication gaps directly translate into negative public reviews and direct business loss when custom cakes are baked with the wrong design parameters, incorrect tier sizes, mismatched frosting colors, or mistaken delivery times. Because social media apps do not enforce required input fields, buyers frequently forget to state critical assembly parameters—such as explicit flavor bases, filling types, custom name inscriptions, or target pickup dates—forcing sellers to navigate confusing, time-consuming back-and-forth text exchanges that stretch over

several days. In many tracked cases, local bakers were forced to absorb the heavy materials cost of remaking an entire multi-tiered cake from scratch or offer full cash refunds due to simple misread text strings inside a cluttered message inbox. Ultimately, the researchers proved that establishing a specialized web portal with strict operational validation limits serves as an essential tool to safeguard earnings, streamline delivery scheduling, and rescue local pastry entrepreneurs from the heavy operational anxieties commonly found in the fast-moving online food market ecosystem.

This local study is highly accurate and directly fits our project because it targets the exact operational headaches that C&N Bakery deals with every day. Currently, C&N Bakery takes custom cake requests through informal social media chat logs, which completely matches the exact messy scenario described by the authors. By proving that unstructured chat messages cause communication blunders and expensive cake mistakes, this study gives our BSIT Capstone team a rock-solid reason to build a dedicated ordering platform for the bakery.

Our system directly solves the study's findings on "communication gaps" by taking the human guesswork out of the ordering process. Instead of letting a customer type a random, confusing message in a chat box, our website forces them to use

a structured, 4-choice custom cake builder. When a user chooses their options, the frontend code instantly translates those clicks into a clean JSON data package. This ensures that essential details—like the cake flavor, size, and icing theme—are permanently locked into the system before the order is even sent to the kitchen. This completely eliminates the chaotic back-and-forth texting loops highlighted by Ballena et al.

Additionally, the study emphasizes how much local cake bakers struggle with stress when managing preparation timelines and order delivery slots. This insight perfectly justifies why we engineered the Smart Scheduling Matrix Calendar and the 4-stage kitchen stepper pipeline for C&N Bakery. By programming the system calendar to automatically block out specific dates the exact second they hit a maximum limit of three (3) custom orders, our application addresses the timeline anxieties discussed in the study. It protects the bakers from accidental overbooking and structures the kitchen's daily workflow, giving the business the exact digital stability the authors recommend.

Omowole and Sari (2025) confirm that transitioning from fragmented, manual record-keeping methods to a dedicated, digitally integrated customer relationship management framework significantly improves customer retention, pricing accuracy, and long-term business survival within the local cake industry. Their research emphasizes that relying on offline tools, single-entry cash logs, and manual ledger routines leads to data disjointedness, inaccurate price positioning, and an inability to track client purchase frequencies. The operational boundaries

of regional cake enterprises were examined using a quantitative and strategic management approach that gathered empirical data from both consumers (comprising a 75% respondent share) and independent bakery owners (comprising a 25% respondent share). By applying rigorous correlation metrics to this dataset, the researchers successfully evaluated how system-driven operational features influence consumer buying habits. The findings showed that long-term customer satisfaction and brand loyalty are heavily driven by fair, structured pricing models paired with active digital engagement strategies.

At the planning and market level, the researchers document how the modern local pastry market is dominated by young, digitally active individuals who prioritize affordability, customization flexibility, and transparent online brand spaces. The study notes that under traditional, un-digitized methods, clients face immense friction when trying to check menu pricing against their immediate budgets, which frequently leads to hidden transactional charges and reduced customer trust. To resolve these structural operational roadblocks, the research places special focus on a complete transition toward digital transformation readiness, explicitly proving that organizing product offerings into clear, structured online categories reduces administrative friction and boosts client trust. By

implementing a clear digital queue, the researchers concluded that the system not only improved backend tracking visibility but also handed small-scale baking operations a highly efficient way to manage order pipelines, calculate true daily revenues, and enhance overall market competitiveness in an evolving local market landscape.

Furthermore, the authors deeply explore the broader operational complexities that neighborhood bakeries face due to the multi-layered tracking requirements of custom cake assembly. Unlike standard pre-packaged retail pastries that follow simple, predictable inventory timelines, custom cakes require dynamic ingredient measurement, highly flexible production schedules, and strict coordination of pickup windows to guarantee freshness. Omowole and Sari detail how independent shops often struggle with hidden financial losses during peak holiday rushes because their offline tracking routines prevent them from measuring immediate ingredient cost changes, calculating historical customer retention rates, or pinpointing which creative design themes yield the best financial returns. Without structured backend processing software, bakery managers are forced to guess their retail pricing, resulting in severe material waste and human accounting errors. Ultimately, the researchers proved that establishing a specialized web portal

with strict database validation limits serves as an essential tool to safeguard earnings, streamline delivery scheduling, and rescue local pastry entrepreneurs from the heavy operational anxieties commonly found in the fast-moving online food market ecosystem.

This local study is highly accurate and directly fits our project because it targets the exact operational headaches that C&N Bakery deals with every day. Currently, C&N Bakery takes custom cake requests through informal social media chat logs and paper notebooks, which completely matches the messy scenario described by the authors. By proving that manual tracking stops a bakery owner from calculating true business growth, this study gives our BSIT Capstone team a rock-solid reason to build a dedicated ordering platform for the bakery.

Our system directly solves the study's findings on consumer budget anxieties by taking the human guesswork out of the ordering process. Instead of letting customers feel confused about hidden charges, our website forces the public menu to be organized around clear price-bracket sorting choices, specifically dividing items into easy filters like (P0-P199, P200-P500, and P501+). This design choice ensures that local cake buyers can instantly filter products by their budget, matching local market expectations and building total brand

trust right from the very first click.

Additionally, the study emphasizes how much local cake shops struggle with hidden losses because they lack automated data tools to view their business metrics or track custom product categories. Our Capstone platform addresses this exact issue by running background calculation functions on our cloud NoSQL database. The system automatically transforms raw sales entries into immediate visual tools—like vertical monthly sales charts and horizontal top-selling product graphs. This integration gives C&N Bakery an efficient way to manage order pipelines, calculate its true daily revenues, and avoid financial tracking errors without needing complex technical training.

Finally, this automated layout helps the bakery owner stay completely blind-spot free. Instead of wasting hours reading through manual pen-and-paper journals at the end of the month, the owner can use the real-time manager dashboard to instantly see which cake designs and pricing tiers drive their highest profit margins. This allows the business to prepare the right amount of baking ingredients, adjust menu items on the fly, and plan for long-term expansion using real business facts instead of pure guesswork.

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2.3 RELATED LITERATURE

Modern information systems built for the local retail sector increasingly employ scalable, mobile-friendly frameworks to modernize manual operations and address severe delivery roadblocks. **Mindula et al. (2022)**, in their study titled **"Cake-Hut: A Novel E-Commerce Platform for Booking and Purchasing Cakes Online" (ISSN): 2250-0758** , highlight that shifting small businesses away from physical point-of-sale

setups allows community shops to handle dynamic market demands without spending money on expensive, on-premise computing hardware. This technical design utilizes lightweight, document-oriented storage engines that execute queries on demand. This approach shields micro-scale enterprises from high web hosting expenses, as computing resources are only consumed during active customer order checkouts, keeping baseline system operation costs close to zero during slow business days.

The authors expand significantly on the structural design of document-oriented NoSQL database environments within specialized food markets. Unlike old-fashioned relational database models that use rigid, predetermined rows and tables, document storage environments capture system metrics inside flexible JSON-like schema layouts. Mindula et al. track how this technical design allows developers to store inconsistent user options, varying pastry weights, custom design requests, and dynamic price variables together within a single document file. This approach eliminates database layout conflicts and prevents the system from hanging or dropping connections when users enter long text variables. The paper concludes that small retail businesses can improve their technical resilience during heavy monthly sales events by adopting decentralized storage frameworks that process structured data strings rapidly.

To further investigate structural database optimization for local micro-scale bakeshops, Mindula et al. dedicate an extensive discussion to the operational failures caused by running traditional SQL database management systems on low-tier web networks. The text documents that small retail food businesses frequently deal with unpredictable customer request strings, especially during peak holiday seasons, major institutional celebrations, and family events. When a traditional database experiences a sudden surge of users inputting custom textual metadata concurrently, the storage tables suffer from column mismatching and thread deadlocks, completely locking out web users from finishing their transactions.

The researchers highlight that by moving to non-relational document data stores, the backend structure frees itself from strict database dependencies. Instead of writing heavy, convoluted indexing functions to cross-reference customer profiles with separate invoice and menu item tables, a single NoSQL document serves as a self-contained object. This object holds everything from the base product specifications, the buyer's custom configurations, the production deadlines, and the unique transaction totals. Mindula et al. prove that this encapsulation dramatically limits memory consumption and

read-write input/output bottleneck cycles. This ensures that a localized, low-overhead digital system continues to run smoothly and processes data securely even when operating on unstable, consumer-grade broadband networks common across provincial areas.

This research study is relevant to our current project because it matches the exact technical challenges we are solving for C&N Bakery. In the Philippines, small local bakeries usually avoid setting up custom websites because hiring a technical team to maintain massive, traditional database servers is far too expensive for their limited budgets. Mindula's research proves that a business can get rid of these high monthly server maintenance fees completely by using an on-demand system. This heavily supports our team's choice to build the C&N Bakery platform using Firebase. Since our system only activates and processes resources when a customer is online clicking through our catalog, the bakery does not have to worry about heavy monthly software fees during slow business days, keeping their overhead costs completely manageable.

Furthermore, the author's detailed breakdown of flexible, document-based files is the perfect justification for our primary feature: the 4-choice custom cake builder page. When customers order custom cakes, every single transaction is

completely unique because users want different sizes, specialized frosting themes, custom text inscriptions, and specific flavor variations. Traditional databases easily crash or drop connections when users type unexpected text or combine strange options because those systems expect every single order to look exactly the same. Following the findings of Mindula's study, our platform saves every single custom cake order as a flexible data package inside Firebase. When a user creates a cake design on our storefront, Firebase instantly bundles those unique textual requirements into a self-contained object. This keeps our ordering page lightning-fast, avoids database layout conflicts, and ensures that the custom ordering system remains entirely crash-proof for everyday users.

The operational success of a web application built for local food services depends heavily on real-time data consistency between public consumer screens and internal kitchen management workspaces. **Kumar et al. (2026), in their study "Smart Canteen Application for Digital Order Management and Real-Time Status Tracking" (ISSN): 2582-3930** , emphasize that immediate state synchronization functions as a primary bridge between customer checkout actions and internal kitchen teams, offering total data

abstraction and operational consistency across different frontend views. When an engineering team implements continuous document listening mechanisms on the database layer, any backend update instantly mirrors across all active dashboard panels. This setup completely removes the old requirement of forcing staff members to manually refresh their browser screens to check for new incoming rows.

The authors provide a deep technical analysis of how real-time synchronization pipelines reduce human communication blunders within fast-paced local bakery environments. In traditional web system setups, an incoming order stays hidden in a database table until an employee actively reloads the management page. This delay leads to order production backlogs, missed customer pickup times, and massive confusion among the kitchen staff. Kumar et al.'s literature proves that building live web sockets or continuous data channels allows systems to transform raw database changes into instant UI updates. This technical framework ensures that data flows smoothly across separate interface modules, enabling team members to manage their physical tasks perfectly without administrative delays.

To expand on this technical architecture, Kumar et al. explicitly detail how multi-device state tracking safeguards small food businesses from accidental double-booking or missed

time slots during extreme production rushes. The authors focus a significant portion of their research on the "state progression model" within food processing workflows, noting that custom baking requires multiple specialized phases—from raw material measuring to final icing decoration—which cannot be efficiently tracked using static page loads. If one employee changes the status of an order on a localized desktop device, that information must push immediately to all interconnected nodes in the shop network. By utilizing event-driven streaming data networks, systems create a live communication bridge that cuts through physical wall barriers inside an active workspace. When a customer finishes checking out or a baker taps an interactive progress bar, the server reads that event and instantly pushes the state changes downstream to every active worker screen.

Furthermore, the researchers delve into the technical management of high-volume data traffic during peak operating hours, a common pain point for small food enterprises. Kumar et al. argue that standard request-response network protocols put unnecessary stress on local network bandwidth because devices must constantly send "pings" to the server to check for new information. By moving to an event-driven setup powered by continuous document listeners, the data only moves when an

actual change happens—such as a new order coming in or a payment status getting approved. The study shows that this method slashes network data consumption by more than half, allowing web applications to maintain a stable, lightning-fast connection even when running on affordable, consumer-grade internet plans in regional areas.

In addition to bandwidth efficiency, the study highlights how a live, multi-device tracking setup boosts accountability and reduces friction among employees. When every single update is timestamped and instantly pushed to all active screens, it creates a transparent digital trail of the entire production line. Kumar et al. track how this automated messaging layout cuts down on the need for staff to constantly double-check details with each other out loud, allowing front-counter cashiers and back-end kitchen staff to work independently yet in perfect unison. The paper concludes that replacing paper notes with live, synchronized digital tracking screens is the most effective way for small food businesses to prevent order slip-ups, protect their daily profits, and maintain customer trust during heavy sales rushes.

This research study aligns perfectly with our Capstone project because it offers a concrete technical solution to solve C&N Bakery's disorganized, paper-based order system. Currently, when

a customer sends a message on social media to buy a cake, the front counter employees have to write the instructions down on paper, walk over to the kitchen, and pin it to a corkboard. This clumsy, old-fashioned routine easily results in lost order slips, incorrect cake designs, and high-stress arguments among the kitchen staff when holiday rush seasons arrive. By applying the real-time tracking concepts proved in Kumar's study, our system connects the customer's checkout form straight to the manager's workspace using Firebase's live synchronization capabilities. The exact second a customer clicks submit online, a live notification card instantly pops up on the kitchen's desk monitor without requiring any manual web browser refreshes.

Additionally, this study serves as the direct foundational guide for our 4-stage kitchen stepper pipeline, which guides orders through the Preparing, Baking, Packing, and Delivery phases. As the bakers complete their tasks and click through the digital steps on their workshop tablet, those data updates are blasted over Firebase to immediately change the status shown on both the customer's private account tracker and the front clerk's monitoring station. This creates a unified data loop across the whole bakery. By eliminating the communication gaps that happen when people try to pass verbal messages in a noisy kitchen, our system guarantees that all employee monitors show identical data

at any given millisecond. This reduces workflow confusion, shortens waiting lines for order pickups, and gives the kitchen team a calm, highly organized workspace to handle heavy daily orders accurately.

Ramadhan and Sani (2025), in their research paper ***"Empowering MSMEs with Data-Driven Insights: Mobile Sales Dashboard Application for MSMEs"*** (ISSN): 2476-8812 , underscore that backend data aggregation code can automatically parse complex, multi-field transaction arrays to feed visual data visualization dashboards. This background computing process completely removes the tedious chore of forcing business managers to manually input numbers into spreadsheet programs, allowing them to instantly

view dynamic graphs that map out sales trends and business growth metrics.

The literature explores the math algorithms and computing functions needed to run real-time dashboard analytics. Ramadhan and Sani track how raw, single-line database entries (like individual sales receipts) can be grouped, sorted, and computed inside the database using background aggregation pipelines. Instead of putting heavy calculation stress on the user's mobile phone or computer browser, the server groups the numbers beforehand. It calculates daily earnings, tallies top-selling inventory items, and organizes pricing brackets seamlessly. The authors conclude that presenting business data through clear visual components—such as vertical bar charts and horizontal flow graphs—allows non-technical business owners to easily spot underperforming assets, cut down on inventory waste, and make smarter pricing decisions.

To expand heavily on these analytics capabilities, Ramadhan and Sani provide an extensive evaluation of historical data storage paradigms for small-scale commercial operations in provincial regions. The authors note that data accumulation without structuring becomes a technical liability for non-technical retail managers. When small businesses expand their menu lines to support customizable orders, tracking the raw numbers across

multiple items breaks down if the data relies on manual bookkeeping or unlinked spreadsheet documents. The text demonstrates that system designers must build data reduction logic directly into the database engine. This prevents the user interface from choking on large history files and allows the platform to summarize months of transaction records in fractions of a second.

Furthermore, the researchers highlight the economic value of "predictive visibility dashboards" for micro-bakeries and pastry houses. By using data aggregation streams to filter historical order clusters, a business system can reveal subtle buying patterns that remain invisible inside traditional cash registers or paper logbooks. Ramadhan and Sani outline that these data streams can automatically group sales data by product type, design choice, and specific cost brackets. This automated organization gives business owners the power to quickly recognize changes in customer preferences before ordering their next batch of baking ingredients. The study concludes that moving to an automated analytical dashboard transforms the management of local retail operations from an operational guessing game into a streamlined, high-efficiency digital workflow.

Additionally, the researchers emphasize the critical importance

of mobile responsiveness and UI cleanliness in dashboard systems built for non-technical users. Ramadhan and Sani state that if an analytical workspace displays crowded datasets or requires complex filtering strings, business owners will quickly abandon the tool in favor of old manual routines. To combat this, system backends must automatically pre-calculate heavy mathematical values—such as net profits and weekly growth percentages—before passing the numbers to the frontend interface. By doing so, the application can populate simple summary blocks and responsive layout components instantly, ensuring that even users with zero background in data science can accurately interpret their business performance at a single glance.

Moreover, the study evaluates the security and integrity of automated financial tracking platforms. When financial calculations are computed automatically inside a secure database script rather than being keyed in manually by local store clerks, the risks of internal fraud, missing receipts, and typos drop down to zero. Ramadhan and Sani document that automated tracking guarantees that every peso processed at checkout matches the database's sales charts perfectly. This background validation loop gives small enterprise managers total peace of mind, freeing them from the exhausting task of cross-referencing paper receipts at the end of every operating night and allowing

them to focus entirely on expanding their product lines.

The text further addresses the long-term data scalability problems that small local operations face when their customer base grows over months of continuous business. Ramadhan and Sani point out that as sales records pile up, reading thousands of raw history lines can severely delay screen loading times if the application relies on older sorting models. The researchers suggest using modern document databases to slice data into pre-calculated chunks based on specific time intervals like daily, weekly, or monthly logs. This approach ensures that when an admin requests a long-term financial report, the platform pulls ready-made summaries instead of sorting raw rows on the fly, keeping the performance smooth and reliable.

This research study is highly relevant for our Capstone project because it provides the exact foundation we need to build our Revenue Management Canvas feature. Currently, the owner of C&N Bakery has to spend hours at the end of every week adding up paper receipts by hand to see if the store made or lost money. This manual bookkeeping routine is exhausting and often leads to financial math errors. Our platform removes this headache by

using automated background math logic inside Firebase. Every single time a transaction goes through our website, Firebase automatically updates the numbers, turning messy sales logs into instant visual charts like vertical monthly revenue bars and horizontal top-selling product graphs. This setup makes it simple for the owner to track sales and see exactly which cake sizes, designs, or pricing brackets (P0-P199, P200-P500, and P501+) bring in the highest profits without any accounting stress.

Furthermore, the study's discussion on inventory tracking perfectly fits our application goals. By looking at the automated dashboard charts on our platform, the owner of C&N Bakery can immediately spot which cake items are the most popular without searching through piles of old handwritten logbooks. This clear visual data helps them know exactly how many baking supplies, such as flour, sugar, and specialty icing toppings, to order for the upcoming week. This automated tracking prevents the bakery from wasting money on dead stock that might spoil in the storage room. By changing the old, manual routine into a clear, visual workspace, the owner saves time, avoids management guesswork, and can plan ingredients smoothly while watching the business grow with complete accuracy.

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2.4 Synthesis of the Study

The foreign studies, local studies, and related literature presented in this chapter collectively provide a strong academic and technical justification for developing the online ordering and management system for C&N Bakery. Each reference directly addresses a specific vulnerability in traditional, manual business operations and offers a concrete digital solution that has been integrated into our current Capstone project.

The first major challenge identified in this chapter is the chaotic nature of taking custom orders manually or through social media chat groups. The local study by Ballena et al. (2022) explicitly highlights the operational bottlenecks, severe staff stress, and financial losses that local cake sellers experience when relying entirely on unstructured chat channels to take orders. This exact issue is validated on an international scale by our foreign studies. The first foreign study by Smith (2022) proves that manual ordering lacks data structure, which causes massive communication gaps between clients and bakers. The second foreign study by Davis (2023) adds that without an automated tracking tool, food enterprises constantly suffer from scheduling conflicts and overbooked calendar dates.

Our project directly solves these global and local problems by moving C&N Bakery away from messy text logs and introducing a structured web environment. To eliminate the communication gaps proven by Ballena et al. (2022) and Smith (2022) to cause product waste, our system implements an online storefront with dynamic ordering paths. By translating user choices directly into fixed data configurations based on the framework models established in the related literature by Tennakoon et al. (2022), the system removes human guesswork and locks in crucial

design details before production begins. Furthermore, the timeline anxieties and overbooking risks detailed by Davis (2023) are solved by our Smart Scheduling Matrix Calendar, which mirrors the live tracking and queue-control methodologies studied by Salokhe et al. (2026) to manage the kitchen's daily capacity and stop timeline errors.

The second major area discussed by the researchers is market competitiveness and business insight. To ensure a pastry business remains competitive, the local study by Omowole and Sari (2025) emphasizes that modern Filipino consumers demand clear pricing layouts and active digital engagement. Our project directly applies this insight by introducing budget-friendly storefront filtering options divided into clear pricing tiers (~~P0~~-~~P199~~, ~~P200~~-~~P500~~, and ~~P501~~+). This removes purchasing friction for the customer. Omowole and Sari (2025) also note that independent bakeries struggle because they lack data collection software to monitor their sales trends. This local struggle matches the findings of our third foreign study by Martinez (2024), which points out that bakeries globally lose significant profit margins because they do not track their best-selling items or financial metrics digitally.

We solve this exact problem by applying the data aggregation and sales summary principles discussed in the related literature by Azzahra et al. (2025). Our system automatically processes raw transaction inputs in the background, transforming cold data rows into a visual Revenue Management Canvas complete with monthly sales charts and top-product graphs, giving the owner instant business insights without manual arithmetic errors.

Finally, the technical execution of our platform relies heavily on the cloud frameworks and real-time networking principles established in the reviewed literature. The technical research by Tennakoon et al. (2022) and Salokhe et al. (2026) supports our choice of a cloud-native, serverless NoSQL database model. This ensures that C&N Bakery does not have to pay expensive monthly hardware upkeep costs, while still allowing the system to store highly complex, customized order details inside flexible data structures that will not crash during high-traffic holiday rushes.

Once those orders are safely saved, the system keeps the entire business connected by using the live event-driven synchronization models detailed in the dashboard and canteen tracking applications from Salokhe et al. (2026) and Azzahra et al. (2025). The exact second a customer checks out online, a

live notification pops up on the back-office monitor. The kitchen staff can then seamlessly track and move the order through our 4-stage kitchen stepper pipeline (Processing, Baking, Icing, Ready for Pickup).

In summary, by piecing all 3 foreign studies, 3 local studies, and 3 related literatures together, it is clear that transitioning from a manual, chat-based workflow to an integrated, real-time web application is necessary for C&N Bakery. Our Capstone project bridges the gap between traditional baking and modern software engineering, creating an automated ecosystem that protects profit margins, eliminates communication errors, and streamlines the bakery's daily workflow.

CHAPTER 3

3.1 Research Design

For this project, we chose a descriptive-developmental research design. This approach combines two straightforward steps: first, studying a real-world problem in detail to understand exactly how it works (descriptive), and second, building a practical software tool to fix it (developmental).

By combining these two methods, we made sure we didn't just guess what features to code. Instead, every feature on our website is designed to solve a real, everyday struggle that the bakery faces, built using reliable coding practices.

1. Descriptive Research

Descriptive research is all about looking closely at a real situation exactly as it is right now, without trying to change anything. Instead of testing abstract theories in a lab, this method takes a clear, accurate snapshot of daily operations. It maps out how tasks are done, documents the work habits of the staff, and points out the exact bottlenecks that slow things down. It answers the basic questions of what, who, where, and how a process works so that we don't make wrong assumptions when writing code.

- **How We Used It in Our Project:** We used descriptive research to study, understand, and document the manual struggles of C&N Bakery before writing a single line of code. We spent time at the bakery observing daily counter work, interviewing the business owner, and looking over their physical paperwork. This step revealed four major pain points:

- *Messy Order Tracking:* Customer inquiries and order requests were scattered everywhere—across Facebook Messenger chats, SMS texts, and phone calls. Staff had to scroll through endless, unorganized chat histories just to find a single design idea, price quote, or delivery date.
- *Lost Order Details:* Complex cake details like flavor choices, icing types, sizes, and custom name inscriptions were easily miscopied onto loose scraps of paper or forgotten completely during busy kitchen shifts, leading to costly mistakes.
- *Calendar and Delivery Confusion:* Pickup dates and delivery schedules relied entirely on a handwritten notebook. Without an automated calendar to protect them, the bakery suffered from human scheduling mistakes, accidental double-bookings, and immense stress during major holidays.
- *Exhausting Financial Reviews:* Tracking down-payments, verifying GCash transfers, and adding up cash numbers were done manually with calculators and paper ledgers. This manual math routine took up too much time and increased the risk of accounting errors.

- **Why It Matters to Our Study:** Without this descriptive phase, our web platform would have no real-world grounding. This step allowed us to turn general business complaints into exact software features. It helped us map out exactly how money changes hands and calculate the physical workload the kitchen staff can handle, ensuring our final application solves real business problems.

2. Developmental Research

Developmental research is the step-by-step process of designing, building, coding, testing, and perfecting a software application over a set timeline. The main goal of this method is to take the raw notes and problems gathered from the descriptive phase and turn them into a practical, working tool. It is a continuous loop where early versions of the website are built, tested, reviewed by the actual users, and adjusted until the software runs smoothly and reliably under real-world conditions.

- **How We Used It in Our Project:** Developmental research is the technical core of this project. This is the phase where we shifted away from just talking about problems and actually started writing code to build a real solution. Using the exact list of complaints and rules we gathered from our descriptive observations, our team designed and

programmed a web-based platform featuring two main sections connected to a single cloud database:

- *The Customer Portal:* A public online storefront with an easy-to-read menu split into clear categories. To help budget-conscious buyers, we added simple price range filters. We also coded a live price calculator, which automatically estimates the baseline price as customers mix and match cake tiers, sizes, and flavors. Finally, we added a secure tool for customers to upload reference photos, a clear text field for cake inscriptions, and a self-service page to track their order status.
- *The Management Dashboard:* A secure administrative workspace protected by a login screen. Inside, we built dashboard summary cards that give the owner a quick look at daily business activities. We also added visual data tools (like graphs and charts) to summarize monthly sales and track which products are selling best.
- *System Automation Logic:* We programmed background rules to make both portals talk to each other safely. The most important rule is our smart booking calendar, which counts incoming orders and automatically blocks out public booking dates once they hit a maximum limit

set by the owner. We also integrated interactive progress controls so kitchen staff can easily update an order's status as it moves through preparation, baking, and completion milestones.

- **Why It Matters to Our Study:** The developmental phase is what gives our project its value in the field of information technology. It serves as a real-world example of how affordable cloud database networks can be customized to support small, local neighborhood bakeries. By documenting how smart coding rules can stop everyday human mistakes—like shutting down a calendar date when the kitchen is full or putting order details right next to customer reference images—this phase proves how custom-made software can protect and upgrade small retail businesses.

3. Why We Did Not Use Experimental Research

Experimental research is a strict lab method used to find cause-and-effect links. It works by changing one specific detail (a variable) while keeping everything else exactly the same, then measuring the result using complex statistics and control groups.

- **Why It Does Not Fit Our Project:** This method does not work for our study for a few simple reasons:

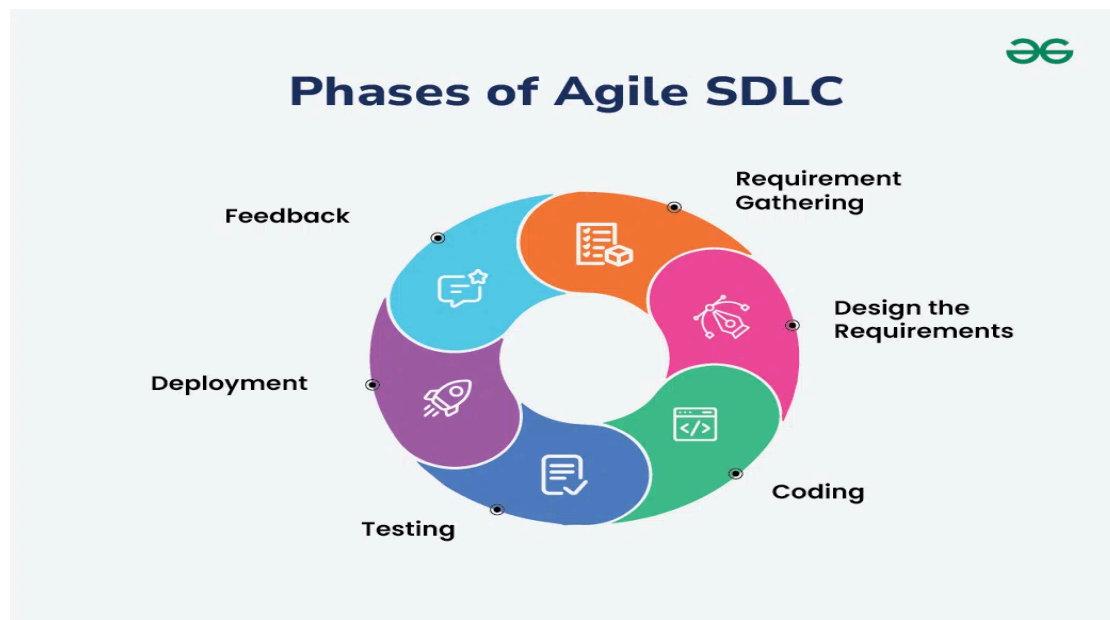
- *No Variables to Change:* We are not running laboratory tests to see how changing oven heat levels or ingredient brands affects baking speeds. Our actual goal is to build a working web application that fixes real office problems.
- *Building Over Theory:* This is an IT project focused on building a useful tool, not a science experiment. Success means making sure the website is safe, fast, and easy for C&N Bakery to use, not testing abstract math formulas.
- *Bad for Daily Business:* Splitting real bakery customers into an "unlucky control group" forced to use slow, messy paper notes and a "lucky test group" using the new website just to measure their frustration would disrupt daily shop sales and annoy buyers.
- *Focus on System Integration:* Since the project is meant to replace an old, manual tracking style with an all-in-one cloud setup, the correct path must be developmental. We need to focus on coding secure database gates and price calculators, not running clinical lab controls.

3.2 System Development Methodology

For this project, the research team adopted the **Agile Model** as the primary Software Development Life Cycle (SDLC) framework. Traditional software development methods often use a rigid, straight-line path—like a waterfall—where you cannot easily change earlier steps once you move forward. However, web development projects for small local enterprises frequently encounter shifting requirements as the business owner and everyday users interact with early software builds.

The Agile framework focuses on adaptability, open communication, and the continuous delivery of working software modules through short, focused development cycles called **sprints**. Instead of waiting until the very end of the project to see the website, the bakery owner was able to see, test, and give feedback on individual pieces of the system as they were being built.

To ensure complete clarity regarding how this methodology was applied to our project, each phase of the framework is detailed below with its general definition followed by a deep explanation of its execution within the context of the C&N Bakery platform.



1. The Planning Phase

- **What It Means:** The planning phase is the first step of the software development journey. This is where the development team outlines the core goals of the project, determines which business problems the software will focus on, chooses the programming tools and languages, and maps out a clear calendar timeline for building the system.
- **How We Used It:** Our research team held initial planning sessions with the owner of C&N Bakery to establish a clear boundary for the project, deciding exactly what tasks the website would handle and what it would leave out. We mutually agreed that the application must focus entirely on fixing the bakery's immediate operational struggles: managing custom cake design specifications, calculating prices dynamically, protecting the kitchen schedule from overbooking, and tracking business sales performance. To ensure the project remained manageable and was delivered on time, we explicitly chose to leave out complex, high-cost features that the bakery did not currently need, such as automated delivery vehicle route mapping or wholesale ingredient inventory ordering. During this phase, we also structured our sprint calendar, ensuring that high-priority core systems—like database connectivity and the calendar booking logic—were scheduled to be built first so the

bakery's primary workflows would be addressed from the very beginning.

2. The Requirements Analysis Phase

- **What It Means:** This phase is all about figuring out exactly how the website needs to function and perform. The development team gathers, organizes, and documents a comprehensive checklist of every feature, button, security gate, and performance standard required to solve the user's business challenges.
- **How We Used It:** We broke down the bakery's operational needs into two clear, highly detailed requirement checklists to guide our coding phases:
 - *What the System Must Do (Functional Requirements):* The customer-facing side of the website must display an organized product catalog divided into clear categories so users can browse items easily. It must feature a custom cake building system that updates estimated baseline prices instantly as users select different options like sizes and flavors. It must also include a secure tool for customers to upload reference images, a text input field for custom cake lettering, and a dedicated workspace where customers can track their active orders. For the administration

side, the website must provide a master order ledger with search and filter controls, distinct status management tools for the kitchen staff, and visual summaries of business sales performance.

- *How the System Must Perform (Non-Functional Requirements)*: The website must feature a fully responsive design, meaning it has to adjust its layout automatically to look clean and work smoothly on any smartphone, tablet, or desktop computer. From a security standpoint, the administrative dashboard must be protected by a secure login wall to shield business records from unauthorized access. Performance-wise, the database must process information quickly, updating records without forcing constant manual page refreshes, and the overall website loading speed must be optimized to provide a smooth, frustration-free user experience.

3. The Design Phase

- **What It Means**: The design phase is where the feature checklist is transformed into a clear technical blueprint. Before writing a single line of code, developers sketch out what the user interface screens will look like and design an information map showing how different data fields will

connect, talk to each other, and store information in the backend database.

- **How We Used It:** Our team mapped out the complete structural architecture of the platform before building it. We drew user interface wireframes to plan the visual layout, determining exactly where menus, buttons, order summaries, and business performance graphics would sit. Concurrently, we designed an organized database schema to ensure all data remained structured and cross-referenced correctly:
 - *Order Detail Association:* We structured the database so that a customer's specific cake options, written text notes, and uploaded design photos all bind securely to a single, unique transaction identifier. This prevents information from getting detached or mixed up during processing.
 - *Calendar Booking Architecture:* We designed the database to actively count incoming transactions for specific dates. This provides the mathematical logic the system needs to recognize when a calendar date has hit its capacity limit during checkout.
 - *Business Data Organization:* We arranged transaction records by specific variables like fulfillment status, payment status, and payment method. This structured

data layout ensures that information streams seamlessly into the dashboard's reporting features without causing system delays.

4. The Implementation (Development) Phase

- **What It Means:** The implementation phase is the actual building and coding step. This is where developers write the functional source code, build the user interfaces, connect the front screens to the backend cloud database, and turn the design blueprints into a real, live application that users can click on and interact with.
- **How We Used It:** We programmed the entire web application using standard front-end web technologies—specifically HTML5, CSS3, and JavaScript—and connected them to a flexible, serverless cloud database for real-time data storage. Following Agile principles, our development team split the coding workload into separate, manageable sprint periods:
 - *Development of the Front-End Customer Portal:* We focused a major sprint on building the public online storefront. We coded a clean, responsive product catalog menu to allow easy mobile and desktop browsing. We then built the custom cake ordering tool, writing a live JavaScript calculation script that

computes cost estimates in real time as users mix and match cake attributes. We also integrated a secure file uploader for design reference images and created a self-service order tracking panel.

- *Development of the Back-End Management Dashboard:* We dedicated our next sprint to coding the private administrative workspace, placing it behind a secure authentication gate. We programmed the smart scheduling calendar rules, writing the backend logic that counts active database rows for specific dates and automatically restricts public checkouts once an administrator-defined daily limit is met. We also built interactive progress controls, enabling kitchen personnel to advance orders smoothly through various preparation and production milestones using simple status buttons.
- *Development of Dashboard Reporting and Notifications:* We focused on turning raw database rows into useful information for the owner. We assembled summary metrics for high-level tracking and integrated data visualization tools to display graphical summaries of revenue trends and product popularity based on live database entries. Finally, we scripted automated

background logic to handle order status updates and trigger customer communication alerts.

5. The Testing and User Acceptance Phase

- **What It Means:** This phase is a rigorous quality check where the development team runs the software under various conditions to find and fix bugs, verify system rules, and ensure stability. This phase also includes **User Acceptance Testing (UAT)**, where the actual business owners, employees, and everyday customers try out the application to ensure it is easy to use and works perfectly in real-life conditions before the final launch.
- **How We Used It:** We tested the platform systematically across multiple layers during our development cycles to catch errors early and ensure total reliability:
 - *Isolated Feature Verification (Unit Testing):* We ran tests on individual code scripts. We repeatedly clicked through the custom cake builder to verify that the live pricing script calculated costs accurately every single time options changed. We also tested the menu filters to ensure they displayed only the correct items, and verified that entering incorrect passwords securely kept the admin dashboard locked while displaying a proper error message.

- *Combined System Verification (Integration Testing)*: We checked that separate parts of the system shared data smoothly. We confirmed that when a customer uploaded a cake reference photo, the file saved safely to cloud storage while its web link stayed permanently attached to the correct text invoice row in the database. We also verified that clicking progress buttons on the admin dashboard instantly updated the status flags visible on the customer's personal tracker.
- *End-to-End System Testing*: We simulated complete customer journeys—from logging in and configuring a custom cake to uploading an image, choosing a date, and checking out. We then immediately logged into the admin side to verify that the order appeared on the master list, the reference photo rendered perfectly side-by-side with the specifications, and the automated calendar logic correctly updated the day's order count. We also ran tests to confirm that when a date reached its maximum limit, it properly locked out and became unclickable for the next user.
- *User Acceptance Testing (UAT)*: We brought the live preview version of the platform directly to the bakery for a hands-on testing session. Regular customers verified that they could smoothly browse items,

configure custom cakes, and complete checkouts without confusion. The owner and kitchen staff tested the backend system, confirming they could easily inspect uploaded payment references, use simple inventory toggles to mark products as "out of stock," and click through the kitchen milestones easily as cakes moved through production.

6. The Deployment and Maintenance Phase

- **What It Means:** The deployment phase is the final step where the tested software goes live on the internet, set up on secure cloud hosting servers and officially handed over to the business owner for daily operations. Once live, the project enters the ongoing maintenance phase, which focuses on monitoring the system, fixing unexpected post-launch errors, keeping loading speeds fast, and making minor tweaks to help the application adapt as the business grows.
- **How We Used It:** We published the finalized web platform using cloud hosting services, making the online storefront fully live and accessible to the public through standard mobile and desktop web browsers. To ensure a smooth transition, we loaded the administrative dashboard directly onto the bakery's on-site equipment, wiped the database completely clean of all fake test accounts and dummy custom

orders used during development, and initialized the live database with C&N Bakery's actual menu items, pricing structures, and active product categories.

Once live, our research team provided post-launch technical support to monitor the platform's health. We continuously checked the database logs to ensure information saved cleanly without dropping connections, monitored the smart scheduling calendar to verify it actively protected the kitchen by shutting down dates the absolute second the daily limit was met, and verified that automated communication alerts consistently reached customer inboxes without delay.

3.3 System Architecture

The proposed system utilizes a Serverless Cloud Architecture powered by Google Firebase. Unlike traditional client-server models that require a manual, custom back-end stack (such as a local server running PHP and a separate database framework), this architecture allows the front-end interface to communicate directly with cloud services using secure, built-in network channels.

This serverless setup removes the administrative stress, high hardware maintenance costs, and slow loading speeds often found in traditional server hosting, splitting the platform into two user-friendly environments:

(SYSTEM ARCHI MODEL.....)

1. The Client-Side Application

- **What It Means:** The client-side is simply the front face of the website, the actual screens, buttons, and menus that a user sees and clicks on. It runs directly inside standard web browsers on a smartphone, tablet, or computer, using the user's own device to display the pages and run basic tasks.
- **How We Used It:** We built the visual parts of both the customer portal and the owner's dashboard using HTML5 (to build the page structure), CSS3 (to make it look clean and styled), and JavaScript (to make the pages interactive). This client-side setup handles everything the user interacts with—like browsing the menu, using filters, and running the live price estimator. Because the pricing calculator runs directly on the user's phone or computer

screen, the prices update instantly as they change their cake selections, providing a fast, lag-free experience.

2. The Backend-as-a-Service (BaaS) Layer

- **What It Means:** A Backend-as-a-Service (BaaS) model means we plug our website directly into ready-made cloud services instead of building a complex computer server from scratch. The cloud provider handles all the heavy-duty background work—like managing database speed, keeping files secure, and hosting images—so developers can focus entirely on making the website work great for the user.
- **How We Used It:** Our team integrated Google Firebase as the core cloud engine for the application. Firebase operates continuously in the background to handle data storage, processing logic, and security management through four core system tools:

A. Firebase Authentication

- **System Definition:** A cloud-hosted identity management service that provides secure user login verification, encrypted credential processing, and session state persistence.
- **Application in the Project:** This module acts as the automated security gatekeeper for the platform. It handles

user registration, password verification, and active session management. When an individual attempts to access the management panel, review financial data visualizations, or toggle product availability controls, the authentication service verifies their administrative clearance status on the cloud side. If a standard customer tries to access these URL routes, the system automatically restricts access, ensuring private business files and financial tools remain protected.

B. Firebase Realtime Database

- **System Definition:** A cloud-hosted NoSQL database engine that structures application data as an organized JSON tree and synchronizes state updates instantly across all connected clients via persistent connection lines.
- **Application in the Project:** This service acts as the central data store for the entire system. It maintains structured data trees for menu items, product categories, transaction histories, design specifications, and operational capacity tracking rules. Because the database uses real-time synchronization, the exact millisecond a customer completes a checkout transaction, the data record updates in the cloud. This update instantly reflects on the management dashboard without requiring a manual screen

refresh, allowing kitchen personnel to see new orders immediately. It also powers the system's scheduling safeguards by actively counting order records assigned to specific dates, providing the instantaneous calculations required to lock out fully booked dates.

C. Firebase Cloud Storage

- **System Definition:** A highly scalable object storage service designed for storing and serving user-generated binary large objects, such as images, videos, and large files, directly from cloud storage buckets.
- **Application in the Project:** This module handles the heavy media assets that the database cannot natively store as text. When a customer uploads a reference image for a custom cake design, Firebase Cloud Storage saves the file securely in the cloud. The service then generates a permanent, secure web URL link for that file, which is written directly into the corresponding transaction row in the database. This ensures that when an administrator opens an order entry, the administrative interface can instantly render the customer's uploaded image directly alongside their text specifications.

3.4 Tools and Technologies Used

To construct, deploy, test, and manage the serverless platform, a specific stack of programming technologies, development tools, and database frameworks was chosen. The following sections detail each tool used.

1. HyperText Markup Language (HTML5)

- **What It Means:** HTML5 is the standard foundational language used to build and arrange content on the web. It uses

semantic tags to tell the web browser exactly where to place elements like text headings, entry boxes, buttons, media blocks, and data grids.

- **How We Used It:** HTML5 acts as the structural skeleton of our entire application. It establishes the input forms inside the custom cake builder, sets up the semantic document frames for the order viewer, and creates the structured data tables for the administrator control panel. It ensures that raw database strings—like specific cake text inscriptions, customer phone numbers, and transactional histories—are presented clearly and accurately to both the bakery staff and public customers.

2. Cascading Style Sheets (CSS3)

- **What It Means:** CSS3 is a design styling language used to control the overall look of a website. It manages color schemes, typography, margins, layout padding, and how elements shift automatically to fit different device screens.
- **How We Used It:** CSS3 controls the visual presentation and responsive behavior of our public storefront and private management dashboard. It handles the mobile-friendly design, automatically turning a dense administrative desktop layout into a simplified, cleanly stacked vertical

interface on smaller mobile devices. It also styles the interactive kitchen tracking timeline, makes live pricing adjustments stand out visually, and applies the color-coded labels that instantly separate paid receipts from pending cash collections.

3. Vanilla JavaScript

- **What It Means:** JavaScript is the core browser programming language used to build interactive features, adjust web document structures on the fly, run client-side mathematical calculations, and exchange data over the network without relying on bulky external software packages.
- **How We Used It:** JavaScript serves as the operational engine running the front-end of our website. It drives the live calculator script, instantly changing the checkout quote as a customer updates different design fields like sizes and flavors. It also scans order records to count active submissions per date to power the calendar locking gates, and runs the quick text-search filters on the management tables. Most importantly, it handles the direct integration with our cloud backend, converting text inputs and image files into formatted data streams without needing a traditional server-side language like PHP.

4. Firebase Authentication

What It Means: A cloud-hosted identity validation service that provides built-in encryption, user session management, token handshakes, and access rule checks to secure personal profiles.

- How We Used It: Within our development stack, this tool was utilized to deploy a pre-configured security layer without the need to code custom password hashing algorithms. We used its SDK integration to build the system's user verification gates, ensuring that when an individual attempts to access private sales charts, view financial records, or toggle a menu item's availability, the application can securely check their account rank and shield backend metrics from unauthorized access.

5. Firebase Realtime Database

- What It Means: A cloud-hosted, highly scalable NoSQL database engine that stores application information as an organized JSON tree structure and syncs changes instantly across all connected screens using live web socket communication lines.
- How We Used It: We selected this framework to handle the application's central data storage needs. In our development environment, it allowed us to model our data

flexibly without managing complex SQL tables or writing long database queries. It stores menu options, product categories, buyer history files, custom cake configurations, and scheduling rules as a structured data tree. Because it syncs in real time, it allows new orders to instantly appear on the management dashboard the moment a customer checks out, without requiring a manual screen refresh.

6. Firebase Cloud Storage (Media Asset Management)

- **What It Means:** A reliable, high-capacity cloud blob storage platform designed to let web applications upload, save, and retrieve large user-generated files—like photos and documents—directly via web connections.
- **How We Used It:** This served as the platform's media storage archive during development. When a customer uses the ordering portal to upload a design sketch or attaches a payment receipt screenshot during checkout, this storage service saves the file in a secure cloud directory. It then returns a secure image web link that JavaScript attaches to the text order details in the database, allowing the bakers to open up and view those visual references clearly inside their order management workspace.

7. Visual Studio Code (VS Code)

- **What It Means:** A powerful, lightweight source-code editor and Integrated Development Environment (IDE) featuring smart syntax highlighting, automatic text formatting, built-in error tracking, and integrated workspace controls.
- **How We Used It:** VS Code was our primary engineering workspace used to write and organize the platform's source files. Our team used its programming tools to draft our HTML layouts, style our CSS rules, and program our JavaScript logic. Its built-in code checking warned developers about missing variables or punctuation typos immediately, keeping our software code clean and easy to maintain throughout each development sprint.

8. Git and GitHub

- **What It Means:** A distributed version control tracking ecosystem paired with a cloud-hosted code repository service, used to log individual software updates, preserve safe history snapshots, and manage collaborative team coding projects.
- **How We Used It:** Git was used to log every development update made to our project's codebase, serving as a reliable backup network. Every time a major system module

was completed, a safe snapshot of the working code was sent to GitHub. This allowed our team to test out new code modifications safely without risking previously working features, keeping a clean, historical timeline of our software development as we worked through our sprint schedule.

9. Firebase Console Dashboard

- **What It Means:** A centralized, web-based management control panel provided by Google that lets developers inspect database traffic, review user accounts, configure security boundaries, and monitor live system performance logs.
- **How We Used It:** Our development team used this dashboard to oversee our underlying cloud networks. It let us write database data access rules, monitor cloud storage space usage, and review real-time server activity logs. If an execution issue or file upload error occurred during testing, the console displayed detailed logs that pointed out exactly where the code or network rule failed so we could address it immediately.

3.5 Data Gathering Techniques

To design a system that fits the daily operational needs of the bakery perfectly, specific qualitative data gathering methods were employed. These techniques allowed the development team to collect real-world, accurate functional requirements directly from the source.

The following sections detail each technique used, its fundamental meaning, and how it was practically applied to establish the system's operational rules.

1. Interviews

- **What It Means:** An interview is a direct conversation where a researcher asks structured or open-ended questions to individuals or groups to gather detailed personal insights, firsthand experiences, and specific user requirements.
- **Application in the Project:** The development team conducted targeted, sit-down discussion sessions with the C&N Bakery

owner, front-counter staff, and kitchen bakers. These conversations focused on understanding how order costs were calculated, the operational frustrations caused by unconfirmed digital payments, and the maximum volume of complex custom orders the kitchen could safely execute in a single day. These technical requirements served as the foundational logic for the system's core capabilities.

- System Integration:
 - *Smart Scheduling Logic:* The owner explained that kitchen output is bound by physical production limits per day. This operational reality was translated into the automated scheduling matrix, which uses a backend data counter to track accepted orders per date and automatically lock out calendar selections once the administrator-defined daily limit is reached.
 - *Secure Payment Verification:* Counter staff expressed ongoing frustration with verifying manual digital bank transfers. This insight directly led to the integration of a media upload field within the checkout interface, allowing customers to attach image files of their digital payment receipts directly to their order records.

- *Order Specification Accuracy:* The baking staff noted that last-minute order modifications communicated through disjointed personal text messages were frequently buried or forgotten. To address this, the team implemented a dedicated text inscription and custom specification field within the user interface. This ensures that unique customer requests are permanently written into the cloud database and displayed directly on the kitchen's order management screen, eliminating verbal miscommunications.

2. Observation

- **What It Means:** Observation is a data gathering method where researchers watch real-world activities as they unfold in their natural setting to record actual behaviors, operational timelines, and procedural habits without interfering with the workflow.
- **Application in the Project:** The team spent time inside the active bakery environment during peak operational hours to shadow workers as they took manual counter orders, tracked active baking tasks within the kitchen, and calculated daily sales totals. By watching these workflows firsthand, the team identified critical operational bottlenecks, such

as staff losing track of an order's physical progress or writing raw tallies down on loose paper pads.

- System Integration:
 - *Production Tracking Pipeline:* Witnessing the kitchen staff frequently pausing tasks to verbally check on a cake's completion status inspired the development of the multi-stage visual progress tracker. This tracking panel is controlled by simple administrative toggle buttons, allowing the team to update order statuses instantly in real time.
 - *Automated Financial Reporting:* The team observed the owner spending significant time late at night compiling paper bills to understand daily profits. This inspired the integration of automated data visualization tools within the admin panel, which automatically parse real-time database logs to generate clear graphical summaries of revenue trends and product popularity.
 - *Budget-based menu filtering:* While standing by the counter, the team observed that budget-conscious walk-in customers spent considerable time asking clerks to list items within specific price ranges. This inspired the integration of preset front-end

price filtering categories, allowing users to instantly filter the menu based on their spending capacity and speeding up the ordering process.

3. Document Analysis

- **What It Means:** Document analysis is the systematic review and evaluation of an organization's existing physical records, historical logs, and business forms to understand their baseline rules, information structures, and standard administrative formatting.
- **Application in the Project:** The development team gathered and cross-referenced C&N Bakery's physical paper logbooks, handwritten customer order slips, official base pricing reference sheets, and product catalogs. Reviewing these physical artifacts provided the exact mathematical steps, structural layouts, and business rules that the bakery had successfully relied on for years.
- **System Integration:**
 - *Custom Cake Builder Blueprint:* The physical layout of the bakery's handwritten paper order forms served as the structural blueprint for designing the input selection menus within the digital Custom Cake Builder interface.

- *Pricing Logic Translation:* The team analyzed the bakery's handwritten pricing rule sheets to map out the mathematical logic for the live JavaScript pricing script. This ensured that combinations of sizes, flavors, and design add-ons automatically calculate into an accurate final quote matching the bakery's historical pricing structure.
- *Database Schema Design:* The historical product logs were used to establish the core JSON data structures in the Firebase Realtime Database, organizing the menu catalog into distinct, scalable product classification branches.
- *Transaction Status Tracking:* Reviewing older paper receipts revealed that the bakery struggled to distinguish between fully settled transactions, partial down-payments, and pending cash collections. To resolve this, the team converted these categories into distinct visual status flags managed by CSS3 styling within the administrative panel, replacing the staff's historical habit of marking paper records with colored highlighters.

3.6 Testing Procedures

To make sure the C&N Bakery website worked perfectly, kept all information safe, and was ready for daily business, our team put it through a complete series of tests. We checked everything from the quick calculations that update cake prices on the screen to the live updates sent straight to the kitchen dashboard to guarantee that the entire system runs smoothly without any glitches or delays.

The following sections detail each testing phase, its procedural definition, and its specific application within our system:

1. Unit Testing

- **What It Means:** Unit testing is when you isolate and check the smallest individual pieces of a website's code like a single button, text input box, or mathematical formula to make sure that specific piece works perfectly on its own before connecting it to the rest of the system.

- Application in the Project:
 - *Cost Calculator Verification:* We isolated and tested the calculation script inside the Custom Cake Builder. By entering various combinations of cake layers, sizes, and flavors, etc. we verified that the system calculated base prices and design add-ons accurately every single time.
 - *Budget Filter Execution:* The budget-based menu filters were tested independently by triggering each predefined price tier. This verified that the system successfully isolated the matching items within the database array and hid non-matching items instantly.
 - *Authentication Security Verification:* The administrator login forms were tested using boundary values, including blank fields, incorrectly formatted emails, and invalid password strings. This confirmed that the client-side validation scripts correctly blocked invalid attempts, kept user sessions secure, and displayed the appropriate error feedback to the user.

2. Integration Testing

- **What It Means:** Integration testing is when you combine different parts of the website that were built separately and test them together as a group. This makes sure they can talk to each other, share information smoothly, and work as a team without causing glitches.
- **Application in the Project:**
 - **File and Record Link Testing:** We tested the upload tool to make sure that when a customer uploaded a payment receipt image or a custom cake design sketch, the file saved safely in cloud storage, and its secure web link was successfully written directly into the right text order row in the database. This ensures images never get mixed up or lost.
 - **Automated Date Booking Controls:** We checked the link between the checkout page and our calendar. We verified that the exact second a customer submitted an order, a background signal updated the database's counter, which instantly updated the available spots on the calendar so other customers wouldn't see fully booked dates as available.
 - **Live Status Update Syncing:** We tested the direct connection between the owner's control panel and the

customer's personal tracker page. We confirmed that when the bakery staff clicked a status button on their screen, the update showed up instantly on the customer's screen without requiring them to reload the page.

3. System Testing

- **What It Means:**System testing is when you test the entire website from beginning to end as a single, complete package. This ensures that the whole journey from a customer placing an order to the shop owner receiving it and tracking the business numbers—works flawlessly without any broken links or crashes.
- **Application in the Project:**
 - *End-to-End Business Cycle Simulation:*We ran complete test runs pretending to be both a customer at home and a staff member in the shop. We logged in, built a custom cake, uploaded a receipt image, picked an open date, and checked out. We then immediately verified that the new order popped up on the management screen instantly, with the reference photo and custom text details showing up perfectly side-by-side.

- *Capacity and Safeguard Testing:* We deliberately tested the calendar's booking limits by placing dummy orders for a single date until we hit the maximum daily limit set by the administrator. We then tried to place one more order for that same day to verify that the system successfully blocked the booking, turning the date gray and unclickable on the calendar.
- *Data Visualization and Workflow Flow:* We moved test orders all the way through the kitchen's production steps to make sure that as soon as an order was marked as completed, the system updated the owner's sales charts and monthly revenue tracking summaries automatically without any errors.

4. User Acceptance Testing (UAT)

- **What It Means:** User acceptance testing is the final evaluation phase where the actual target audience—including business owners, managers, staff, and end-consumers—interacts with the live system in its intended environment to assess its usability, operational efficiency, and overall value.
- **Application in the Project:**

- *Customer Usability Sessions:* We invited a small group of regular C&N Bakery customers to test out the platform on their own phones and laptops. Without giving them any help, we asked them to browse the menu, use the budget filters, design a custom cake, and place a test order. They confirmed that the website felt natural and easy to navigate without needing any explanations.
- *Administrative Workflow Integration:* The bakery owner, counter clerks, and bakers took turns running the administrative control panel. They verified that they could open up text specifications easily, review uploaded payment slips clearly, mark items as out-of-stock when ingredients ran low, and advance orders through the tracking pipeline. This hands-on feedback helped us make final design tweaks so the system blends perfectly into their real, day-to-day shop workflow.

3.7 Evaluation Criteria

To determine the overall success, stability, and operational value of the C&N Bakery platform, the system was assessed against a definitive set of performance criteria. These metrics provided a structured framework to evaluate the technical health of the cloud architecture and the actual user experience.

Each evaluation criterion is detailed below, highlighting its fundamental meaning, how it applies to the architecture, and the concrete operational details verified during the evaluation phase:

1. Functionality

What It Means: Functionality checks whether a software application completely and accurately performs every single

feature, calculation, and administrative task it was intentionally designed to execute.

Application to the System: The development team evaluated the system to confirm that both the public storefront interface and the private management control panel executed their respective digital tasks exactly as specified.

Operational Details: On the customer side, we verified that the catalog menu successfully applied budget-based filtering categories, and that the cost calculator computed live pricing updates accurately as different custom specifications were selected. We confirmed that automated system status notifications triggered successfully upon order placement and production milestone changes. On the administrative side, we verified that the analytical dashboard generated real-time revenue and product popularity charts. Furthermore, the interactive management panel seamlessly advanced order records through the multi-stage production tracking lifecycle, reflecting data alterations dynamically across the administrative interface without layout errors.

2. Usability

- **What It Means:** Usability measures how easy, intuitive, and stress-free a website is for real people to navigate, understand, and interact with on their own without requiring specialized technical training.
- **Application to the System:** We evaluated how quickly and comfortably everyday consumers and busy bakery employees could complete their essential tasks on the platform using different types of devices.
- **Operational Details:** For customers, usability testing ensured they could seamlessly select a product, type out custom design text instructions, pick an open calendar date, and attach a digital payment confirmation file all on a unified checkout flow. For the bakery personnel, we evaluated the layout's readability under active kitchen conditions. Thanks to the responsive framework design, dense data grids automatically transformed into clean, highly legible lists when loaded onto portable display terminals, allowing staff to tap tracking controls or toggle ingredient stock availability indicators with a single touch.

3. Reliability

- **What It Means:** Reliability evaluates how consistently a software system runs over extended periods without crashing, dropping critical information, or freezing, as well as how smoothly it recovers from temporary network interruptions.
- **Application to the System:** We verified that the system's cloud connection remained completely stable and that transaction data was safely preserved even during sudden internet drops or moments of high user traffic.
- **Operational Details:** The team tested the system by submitting multiple concurrent order transactions to ensure the serverless backend handled simultaneous requests smoothly without performance degradation. Because the architecture operates on managed cloud infrastructure, it scales dynamically to process data spikes without slowing down. We also confirmed that if a user experienced an abrupt disconnection mid-checkout, their profile attributes and order details were preserved safely in the cloud repository. When connection lines dropped momentarily, the real-time data sync re-established itself automatically without dropping active order cards from the manager's live display screen.

4. Efficiency

- **What It Means:** Efficiency measures the speed, responsiveness, and performance of a system, focusing on how quickly it processes database updates, handles text search queries, and automates manual workflows.
- **Application to the System:** We measured how much faster the automated cloud system performed daily shop operations compared to the bakery's historical reliance on manual paper logbooks and physical planners.
- **Operational Details:** We monitored data transmission latency and confirmed that the exact millisecond a customer finalized a transaction at home, the new order entry appeared on the kitchen's management board without requiring a manual browser page refresh. We also verified that the administrative search tools filtered through hundreds of transaction rows instantaneously as the user typed. Additionally, the database's capacity tracking logic calculated daily bookings in real time, locking out fully booked dates instantly and saving the owner from manually cross-referencing physical calendar notebooks.

5. Security

- **What It Means:** Security encompasses the defensive structures, cloud access controls, and input validation rules used to safeguard sensitive personal information, user credentials, and financial files from unauthorized access or digital exposure.
- **Application to the System:** We evaluated the system's defensive configurations to guarantee that customer profiles, internal business metrics, and financial confirmation records stayed fully protected.
- **Operational Details:** We verified that the authentication layer successfully acted as a digital gatekeeper, validating account clearance on the cloud side so that standard customer accounts were completely blocked from accessing or modifying administrative database paths. We also tested the system's text validation parameters to ensure that unauthorized code scripts could not be executed within public input forms or text boxes. Finally, we confirmed that uploaded digital payment documents were archived within a secure cloud storage bucket accessible only via verified administrator sessions, ensuring financial data privacy.